MULTIMODAL COMPUTATIONAL INTELLIGENCE FOR ENHANCING CAREGIVER PERFORMANCE AND CARE DELIVERY

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Individual component: AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

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DECLARATION AND SUPERVISOR CERTIFICATION

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Abstract

This thesis presents AI-assisted caregiver workforce deterioration detection and adaptive workload redistribution, implemented as PULSE, the workforce component of the web-based CareSense platform. Supervisors in care homes often learn that a caregiver is under strain only after it has begun to affect the work. PULSE addresses this by estimating the probability that a caregiver reports elevated stress, using data from wearable devices, and by presenting the estimate to the supervisor together with an explanation and a suggested redistribution of work that a person reviews. The study asked how well a personalized and explainable pipeline ranks elevated stress in caregivers it has not seen before, whether Bluetooth-measured proximity to colleagues adds useful information, and how much the model contributes beyond the caregiver's previous stress response.

The data came from TILES-2018, a study that followed 212 hospital workers for about ten weeks, and they were used under a signed data use agreement. Participants were assigned to training, validation, test and demonstration groups before any feature was built. A first LightGBM model learns the heart rate expected for each person from their recent trend and the day of the week. The gap between the observed and the expected heart rate then enters a second LightGBM classifier, together with fifteen other inputs that were available before each survey.

On 27 held-out caregivers (1,406 assessments), the classifier reached a ROC AUC of 0.7211 (95% confidence interval 0.6644 to 0.7699). Repeated participant-grouped validation gave a mean of 0.6991. Most of the signal came from the previous stress response, and the gain of the full model over that response alone could not be distinguished from zero. Proximity centrality added no detectable value. In a matched test without the previous response, the AUC fell to 0.6108, and later experiments that used validation data did not restore it.

PULSE runs inside a React, FastAPI and Supabase application with role-based access. Each estimate comes with a SHAP-based list of contributing factors, and a rule flag sustained risk and proposes a colleague from the same unit for review. The module is intended as decision support, and a person makes every decision. Its effect on staff stress has not been measured.

Keywords: caregiver stress, wearable sensing, explainable machine learning, personalized baseline, workload redistribution

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The TILES-2018 data set was collected by the Signal Analysis and Interpretation Laboratory at the University of Southern California, in partnership with USC Keck Hospital, and was described by Mundnich et al. [1]. I am grateful to its principal investigator, Dr. Shrikanth Narayanan, to Karel Mundnich and the other authors, and to Ms. Judy Genovese of the USC Stevens Center for Innovation, who signed the data use agreement. The data collection was supported by the Office of the Director of National Intelligence (ODNI), Intelligence Advanced Research Projects Activity (IARPA), under IARPA Contract No. 2017-17042800005. I also acknowledge the 212 hospital workers who volunteered for the study.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application programming interface
AUC	Area under the curve
CEAI	Centre of Excellence for Artificial Intelligence
CI	Confidence interval
CORS	Cross-origin resource sharing
CV	Cross-validation
DSR	Design science research
DUA	Data use agreement
EMA	Ecological momentary assessment
FR	Functional requirement
GNN	Graph neural network
HR	Heart rate
HRV	Heart rate variability
JSON	JavaScript Object Notation
JWT	JSON Web Token
LSTM	Long short-term memory
ML	Machine learning
NFR	Non-functional requirement
PPG	Photoplethysmography
REST	Representational state transfer
RLS	Row-level security
ROC	Receiver operating characteristic
RQ	Research question
SD	Standard deviation
SDG	Sustainable Development Goal
SHAP	Shapley additive explanations
TFT	Temporal fusion transformer
TILES	Tracking Individual Performance with Sensors
UI	User interface
USC	University of Southern California

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1 INTRODUCTION

1.1 Overall Project Context

The group research project is titled *Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery*. It investigates how computational methods applied to several types of data, namely physiological signals, video, speech and sign language, can support caregivers in their daily work and improve the quality of the care they provide. The outcome of the project is a web application named CareSense, which brings four research components together within a single system. Each component was developed by one member of the group, and each member submits an individual thesis on the component that she developed. A separate group thesis describes CareSense as a whole. This document is the individual thesis of Dayana Priyadharshani Kumar. It reports the research and development of the component named PULSE, which addresses the monitoring of caregiver stress and the planning of workload redistribution. The four components and their owners are listed in Table 1.1.

Table 1.1: The four CareSense components and their owners

Component	Owner	Role within CareSense
PULSE	Dayana Priyadharshani Kumar	Stress-risk estimation, personal baseline comparison, trends, explanations and redistribution recommendations for supervisors
SENTRY	Harishalinee Elangovan	Video-based fall detection and alerts
SCRIBE	Vithursha Prabagaran	Conversion of spoken narration into structured records of activities of daily living, with safety alerts
GLOSS	Kaushalya Nagenthraraja	Sign-language recognition and learning feedback

CareSense is intended for use in residential care settings such as care homes and nursing homes. In these settings a small team of caregivers looks after residents continuously, and a supervisor assigns the staff to duties. The data used across the project were not drawn from a single source. They were gathered from several environments, namely care homes visited by the group, volunteers who took part in data collection, and publicly available datasets.

Access to the application is organized by user role. Three roles are defined: caregivers, who carry out daily care, supervisors, who oversee a group of caregivers, and administrators, who manage users and system settings. The four components appear as separate modules inside the same web application, and a signed-in user moves between them from a single navigation menu. A shared audit log records the actions performed by users, for example which person viewed or changed a record and at what time, so that access to sensitive information can be reviewed afterwards.

From a research perspective, the project treats caregiving as a setting in which several kinds of data are produced at the same time but are rarely analyzed together. Each component therefore applies computational methods to one of these data types. PULSE analyses physiological and activity signals from wearable sensors to estimate stress risk. SENTRY analyses video to recognize falls, SCRIBE processes spoken narration to produce structured care records, and GLOSS recognizes sign language to support communication. Because the data types differ, so do the methods, the datasets and the ways in which each model can be evaluated. The research question shared by the group is whether such separately developed models can be combined into one system that is practical for care staff to use.

The research of each component follows the same general approach. A model is first built and evaluated on data that were kept separate from training, and the result is then presented to the user through the application together with information that supports interpretation, such as the factors behind a prediction or a confidence level. This approach reflects a design decision shared across the project: the system provides decision support, and the final decision remains with the caregiver or supervisor. The individual theses report the evaluation of each component, and the group thesis reports how the components were integrated.

1.2 Individual Component Context

AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution is the individual research topic reported in this thesis. The component is named PULSE within the CareSense platform, and the thesis uses that name from this point onward. It addresses a practical difficulty in care work. A supervisor can

usually see that a caregiver is struggling only after the effect has reached the work itself, by which time the options for adjusting that person's duties are limited.

The research is based on the TILES-2018 data set, which was collected from hospital workers by a research team at the University of Southern California [1]. The data were released to this project under a signed data use agreement. They combine wearable sensor records with short self-reported stress ratings given close to the moment of experience, a method known as ecological momentary assessment [2]. PULSE uses the sensor records to estimate how likely a caregiver is to report elevated stress at a given survey. The label used for elevated stress is defined for this research and is not a clinical threshold or a diagnosis. The data and the method are described in Chapter 3.

The estimates are presented in a web interface used by supervisors and administrators. Supervisors see the caregivers assigned to them, and administrators see all monitored caregivers and operate the controls that load and replay historical data. Each estimate is shown with the factors that raised or lowered it and with the history of earlier assessments. When the estimate remains high over several consecutive assessments, the module proposes a colleague from the same unit with a lower current risk as a possible replacement. A supervisor reviews the proposal, and no duty is reassigned automatically. Caregivers do not use the module directly, because showing a stress estimate to the person it describes, without supporting care, could cause harm. The design of the module is described in Chapter 4.

During April 2026 the group consulted care providers about the daily work of caregivers and the information that supervisors need. These consultations clarified the requirements of the module and are recorded in Appendix C.

1.3 Problem Statement

Burnout is generally described as having three parts: feeling emotionally drained, becoming distant from the work and the people being cared for, and losing confidence in one's own ability to do the job well [3]. Maslach and Leiter explain that it develops when a person is exposed to a poorly fitting work situation for a long time. The areas involved are workload, control over one's work, reward, fair treatment and shared

values. Dall'Ora and colleagues reviewed the theories used to explain burnout among nurses. The theories stress different causes, but most of them link burnout to the demands placed on staff and the support and resources available to meet them [4]. Researchers therefore do not fully agree on the exact explanation, yet most accept that staffing and workload play a central part.

The effects are not limited to the staff member. Hall and colleagues reviewed studies on healthcare staff and found that poorer wellbeing and higher burnout were linked with poorer patient safety. They also noted that it was unclear which of the two causes the other, and that the quality of the studies varied [5]. Aiken and colleagues studied more than ten thousand hospital nurses and found that nurses who were responsible for more patients each were more likely to report burnout and dissatisfaction with their jobs, and that surgical patients in those hospitals had a higher death rate [6].

A supervisor who wants to step in early has little information to work with. Many facilities learn how staff are coping only through occasional conversations, or when absences and mistakes begin to appear. Neither source shows how a caregiver's condition changes from one shift to the next. Wearable devices can record body signals all the time, and stress is known to be linked with changes in heart rate and in the variation between heartbeats [7]. A heart rate reading is hard to interpret by itself. Heart rate differs a great deal between people, and it also rises with normal physical activity. A single limit applied to every caregiver would wrongly flag a person whose resting heart rate is naturally high, and it could miss a real change in a person whose heart rate is normally low.

The group interview with the founder of Remark Elder Care on 7th April 2026 showed how this problem appears in a working care home. Two needs were identified. The first was a regular indication of how caregivers are coping, so that strain can be noticed before it affects their work. The second was a practical way to respond, which in this case meant bringing in a substitute caregiver to share the workload.

The problem addressed in this thesis can therefore be stated as follows. Supervisors have no clear way to see, for each caregiver, whether high stress is continuing over time, which factors are behind it, and what a sensible response would be, even though

the signals needed are already being recorded. A tool built for this purpose must state its limits plainly. A stress estimate is an indicator and not a diagnosis, and a suggested change of duties is a proposal for the supervisor to consider and not a decision.

1.4 Research Gap

Studies that detect stress from wearable data have produced useful results, but several questions are still open. The gaps below are the ones this study addresses. Each is stated narrowly, because none of them means that earlier researchers made an error. They describe what has not yet been shown clearly.

- **Timing of the inputs.** Studies that predict a stress rating from sensor data do not always state which readings were available before the rating was given. If data from the whole day are used to predict a response given in the morning, the model appears more accurate than it would be in real use. The timing of each input therefore needs to be stated clearly, and any exception needs to be checked [8], [9].
- **Performance on new people.** A data set usually contains many records from the same person, and these records resemble one another. If the records are divided at random between training and testing, the model can learn to recognize the person and not the pattern of stress. Testing on people who were not part of the training avoids this problem [8]. Even so, such a test shows performance on new people from the same data set and does not show performance in a different workplace.
- **Value of colleague proximity.** It is not clear whether knowing how close a worker stays to colleagues improves the prediction beyond what the worker's own heart rate and activity already provide. This has rarely been tested against a baseline built from the same records. A finding of no improvement is still a useful answer.
- **Comparison with a simple predictor.** A worker's previous stress rating is a strong predictor on its own. A more complex model should be compared with

it, and the comparison should include a measure of uncertainty so that a small difference is not reported as an improvement.

- **Use of personal history.** A model can use each person's own recent pattern to build its inputs without training a separate model for every worker. This gives a personal comparison but is not complete personalization, and the thesis keeps this distinction.
- **Link from explanation to action.** Methods exist that show which inputs contributed to a tree-based model's prediction [10], [11]. It is less common to see these explanations presented together with a suggested staffing action that a supervisor can inspect and review, and to see the basis of that suggestion stated.
- **Use in care settings.** The available data come from hospital workers, and the outcome is a self-reported stress rating, not a measure of burnout. Work on stress monitoring in healthcare has shown that models do not always perform equally well when applied to other data sets [12]. Whether a model built on hospital data can support care home staff is therefore not yet known. This thesis reports the hospital results carefully and identifies this question for further study. It does not claim to answer it.

The aim and objectives in the next section respond to these gaps.

1.5 Aim and Objectives

The main objective of this research is to design and develop a web-based system that uses machine learning to detect early signs of change in the condition of caregivers and to support the adjustment of their workload. The system draws on behavioral, time-based and relational information.

The work reported in this thesis refines this objective in line with the data that were available. The aim is to develop and evaluate a personalized and explainable method that estimates the probability that a caregiver reports elevated stress at a given assessment. The estimate is built from heart rate, physical activity, sleep and the caregiver's previous stress rating. It is personalized because each caregiver is

compared with their own recent pattern and not with the average of all caregivers, and because the method learns what heart rate to expect from that caregiver on a given day. The method does not use the identity of the caregiver as an input, so that it learns patterns and does not simply recognize individuals. It is tested only on caregivers who were not part of training, which shows how it behaves with people it has not seen. The aim also includes testing whether the physical closeness between colleagues improves the estimate, and implementing, within the CareSense web application, a way for supervisors to follow a caregiver's estimated risk over time and to review a suggested redistribution of work.

Six sub-objectives support the main objective.

1. **Workload and behavior features:** Build features that describe the workload and behavior of caregivers from heart rate, activity, sleep and self-reported stress records.
2. **Personal profiles:** Create a personal baseline for each caregiver and measure how far each new reading differs from it, so that changes are judged against that person's usual pattern.
3. **Detection over time:** Follow the estimate across successive assessments, so that a high estimate which continues is distinguished from a single high value.
4. **Relational structure:** Examine whether the relationships between colleagues, measured by Bluetooth proximity, add information about stress.
5. **Explanation:** Show a supervisor which factors raised or lowered each estimate, in a form that can be read without technical knowledge.
6. **Workload redistribution:** Design and demonstrate a rule-based method that proposes a colleague with a lower current risk as a replacement, and leaves the decision to a supervisor.

The way in which each sub-objective was met is reported in Chapters 3 to 6 and brought together in Section 9.2. The research questions that follow organize the analysis of the results.

1.6 Research Questions or Hypotheses

Five research questions guide the analysis in this thesis. The first four concern the research pipeline and ask what the model can and cannot do. The fifth concerns the web application and asks how the results can be presented to a supervisor. Together they cover the main objective of the study and are linked to the sub-objectives in Section 1.5. Table 1.2 states each question and shows where it is answered.

Table 1.2: Research questions and where they are answered

ID	Question	Answered in
RQ1	How well does the final personalized two-stage pipeline separate the assessments in which caregivers reported elevated stress from those in which they did not, when it is tested on hospital caregivers who were not used in training?	Section 6.1.2 and Section 6.1.3
RQ2	Does adding a measure of each caregiver's position in the Bluetooth proximity network improve the estimate beyond heart rate, activity and sleep features, when models with and without the measure are compared on the same records?	Section 6.1.6
RQ3	How much does the full set of features add beyond the previous stress rating on its own, and is the difference large enough to be separated from chance variation?	Section 6.1.4
RQ4	How much does the result change when the steps feature is rebuilt using only information recorded before each survey?	Section 6.1.5
RQ5	How can the explanation of each estimate and the history of repeated high estimates be presented in the web application, so that a supervisor can review a suggested redistribution of work?	Section 4.4, Section 5.2 and Section 6.2

RQ1 concerns overall performance. The estimate is judged by how well it ranks assessments, and the test uses only caregivers who took no part in training, so the result describes how the model behaves with new people. It is reported with a range that shows its uncertainty and is checked against a repeated validation.

RQ2 concerns the relational objective. Models with and without the proximity measure are compared on the same records and in three kinds of model, so any difference can be attributed to the measure and not to a change in the data or the method.

RQ3 concerns the simplest available predictor. A person's previous stress rating is often close to the next one, so the full model is compared with that rating alone, and the difference is reported with its range.

RQ4 checks the timing of the inputs. The steps deviation feature was first built from a baseline that included information from the same period as the survey. The question asks whether the result holds when only earlier information is used.

RQ5 is a design question and is answered by describing and demonstrating the working application. Whether supervisors find the presentation useful has not yet been tested, and this is identified as future work in Section 9.4.

1.7 Scope, Limitations and Assumptions

In scope: The thesis covers the research pipeline and the PULSE web application. The research part includes the allocation of participants into separate groups, the construction of features, the two-stage model, the testing of proximity features, the measurement of uncertainty and the explanation of estimates. The application part includes the backend service, the database with its access rules, and the views used by supervisors and administrators, namely Overview, Risk Analysis, Caregiver Profiles, Trends, Redistribution and Demo Control. It also includes the administrator's controls for uploading and replaying historical readings, and the redistribution logic. This logic proposes a colleague from the same unit with a lower current risk when a caregiver has several recent high estimates, and it leaves the final decision to a supervisor.

Out of scope: No new participants were recruited and no new wearable data were collected. Earlier datasets that were explored during the first progress phase and then set aside are not part of this work. These are a nurse stress dataset collected by Hosseini et al. [13], and the speech emotion datasets RAVDESS [14] and CREMA-D [15]. They were examined as candidates, and the final research uses only TILES-2018 [1]. Clinical diagnosis, burnout classification and any trial of an intervention are also outside the scope.

Assumptions: The five-point stress scale is not an assumption, since it comes from the documentation of the TILES-2018 data set. The assumptions made here concern

how the data are used. A rating of three or above is treated as elevated stress, which is an operational label chosen for this research and is not a clinical threshold. The heart rate and activity records released with TILES-2018 are assumed to be accurate enough for the analysis. The time at which a survey was sent is assumed to be a safe cut-off for deciding which readings could have been known before the answer was given.

Limitations carried forward: This study has two main limitations. First, the outcome that the model estimates is the probability that a caregiver reports elevated stress, and it is based on a self-reported stress rating. It is not a measure of burnout. Second, the participants in the TILES-2018 data set are hospital staff in the United States and not care home staff in Sri Lanka. These limitations are discussed further in Chapter 7.

1.8 Individual Contribution

Everything described in this thesis that relates to PULSE was completed by me, under this research. It covers the research pipeline, the backend service, the database setup, the frontend views and the redistribution work. Table 1.3 summarizes my contribution in each area.

Table 1.3: Individual contribution to PULSE

Area	What I did
Research design	Framed the questions, allocated participants before feature construction, defined the target and the timing contract, and set the evaluation plan.
Data pipeline	Cleaned the survey responses, rebuilt heart-rate features at each survey cut-off, joined activity and sleep summaries, and built the proximity network from Bluetooth records.
Modelling	Developed the two-stage LightGBM pipeline, diagnosed and corrected overfitting, ran the matched centrality experiments and the sensitivity analyses, and computed explanations.
Backend	Wrote the FastAPI service for authorized scoring, history, analytics, raw-data upload, replay and redistribution.
Frontend	Built the React and TypeScript views for overview, risk analysis, caregiver profiles, trends, redistribution and demo control.
Database	Designed the PostgreSQL tables, access rules and migrations for PULSE in Supabase.

Area	What I did
Integration	Connected PULSE to the shared authentication, navigation, audit and model-registry facilities of CareSense.

Beyond the areas listed in the table, my contribution includes the following. I designed the evaluation so that the results can be trusted. The final model was tested on caregivers who took no part in training, and I measured its uncertainty with a caregiver-level bootstrap and a repeated, participant-grouped validation. I also compared the full model with the previous stress rating on its own, and I checked the effect of rebuilding one feature from strictly earlier information. These analyses show how much each part of the pipeline adds. I carried out the matched experiments that tested whether Bluetooth proximity improves the estimate, and a further set of experiments that examined how the model performs without the previous stress rating. They give the project clear evidence on both questions. I connected the model's output to the supervisor's work by adding SHAP factor lists, the rule that flags a caregiver whose estimate stays high over several recent assessments, and the risk tiers that the interface displays. I also gathered the requirements for the module through consultations with care providers, and I documented the research steps so that the results can be reproduced. Together, these parts form a complete line of work, from the data to a reviewable suggestion on a supervisor's screen.

1.9 Thesis Structure

This thesis is organized in nine chapters, followed by the references, a glossary and three appendices.

- **Chapter 1: Introduction.** Presents the project context, the problem, the research gap, the aim and objectives, the research questions, the scope, limitations and assumptions, and my individual contribution.
- **Chapter 2: Literature review and related work.** Reviews studies on burnout and stress in care work, stress detection from wearable data, explainable models and workforce scheduling, and identifies the gaps that this study addresses.

- **Chapter 3: Research methodology.** Describes the research design, the TILES-2018 data, the construction of features, the two-stage model, the proximity experiments, the evaluation measures and the ethical considerations.
- **Chapter 4: Product and solution design.** Sets out the requirements, the architecture of the PULSE module, the data model, the interface and the design of the redistribution rule.
- **Chapter 5: Implementation, integration and testing.** Covers the technology used, the implementation of the research pipeline, the backend and the frontend, the integration with the platform and the testing approach.
- **Chapter 6: Results and validation.** Reports the model results with their uncertainty, the supplementary experiments, the results of the application and a summary of the findings by research question.
- **Chapter 7: Discussion.** Interprets the findings, compares them with related work, and discusses the contributions, limitations and implications.
- **Chapter 8: Commercialization and sustainability.** Describes the value of PULSE to its users, its position among existing options, a possible business model, the questions of scale, ownership and compliance, and the link to the Sustainable Development Goals.
- **Chapter 9: Conclusion and future directions.** Presents the conclusions, the extent to which the objectives were met and the proposed future work.
- **References, glossary and appendices.** Contain the IEEE reference list, the definitions of terms, the allocation of participants and usable data, the disclosure of AI assistance and the consultation records.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The review was designed around the decisions that this thesis has to justify. These are how to define the outcome, how to test a model on people it has not seen, what to compare a stress model against, how to explain a prediction, and how to turn a prediction into something that a supervisor can use. The literature was searched and read with these five questions in mind, and each source was kept only if it informed at least one of them.

The review began with a working bibliography of nineteen candidate papers that I assembled in a research workbook during the earlier phases of the project. It covers work on stress detection from wearable and speech data, burnout among healthcare staff, public datasets, explainable models and the evaluation of prediction models. Literature research was mainly carried out in Google Scholar, IEEE Xplore and PubMed. The search terms combined stress detection, wearable sensors, healthcare workers, burnout, ecological momentary assessment, explainable machine learning, SHAP, nurse rostering, workload allocation. Searches were repeated and extended as the work developed, and the reference lists of relevant papers were also followed. For topics that the workbook did not cover, such as theories of burnout, the design of evaluation studies and workforce scheduling, I added further sources from searches during the writing of the thesis.

Before any claim was taken from a source, I checked the source against its primary record, which means the publisher’s page, the DOI registry entry or the preprint itself. This check changed several entries. A speech emotion paper had been attributed to the wrong journal, a dataset paper carried the wrong year, and a review had been listed under the title of a different paper. Two entries, a graph-based workforce paper and a review of burnout prediction, could not be located under the authors and titles that the workbook gave. I did not use them and drew no argument from them. I also checked what each source actually supports. A review of wearable stress detection shows how a field is organized, but it does not show that a particular design choice is best, so claims were limited to what the source demonstrates.

2.2 Domain and Theoretical Foundations

2.2.1 Stress, burnout and the care workforce

Maslach and Leiter describe burnout through three dimensions: exhaustion, cynicism or detachment, and a reduced sense of professional efficacy [3]. Dall'Ora and colleagues reviewed the theories used to explain burnout in nursing and showed that they stress different mechanisms, from job demands and resources to the match between a person and the workplace [4]. The disagreement matters for a tool like PULSE. A system that claims to measure burnout takes sides in a debate that is not settled, whereas a system that estimates a self-reported stress rating makes a smaller and more defensible claim.

The reasons to care about staff stress are practical. In a systematic review of healthcare staff, Hall and colleagues reported that poorer wellbeing and higher burnout were associated with worse patient safety, although they cautioned about the quality of the evidence [5]. Aiken and colleagues studied hospital nurses and linked each additional patient per nurse to higher burnout, higher job dissatisfaction and higher patient mortality [6]. Workload and wellbeing are therefore connected to outcomes that supervisors are responsible for, and that connection is what motivates giving supervisors earlier visibility.

2.2.2 Measuring stress by self-report and momentary assessment

Ecological momentary assessment collects brief self-reports close to the moment of experience, in ordinary settings and not in a laboratory [2]. Its main benefit is that it reduces the reliance on memory. For modelling, its weakness is that each report is a single answer and can be affected by chance variation. TILES-2018 used this approach. Participants answered one daily survey about job, health or personality, and every version of the survey opened with a question asking them to rate their current stress [1]. PULSE treats that rating as its outcome, turned into a binary label. The label describes how a person said they felt at one moment. It is not a diagnosis and not a burnout score.

2.2.3 Physiological signals and stress

Stress alters autonomic activity, and wearables record some of its traces. Healey and Picard classified the stress level of drivers from physiological signals recorded during real driving, an early demonstration that such signals carry useful information outside the laboratory [16]. Kim and colleagues reviewed the relationship between stress and heart rate variability and found a measurable association [7]. Heart rate variability, however, is computed from beat-to-beat intervals. The consumer wrist trackers behind TILES-2018 report sampled heart-rate values, so PULSE can describe how a person's heart rate is distributed over a period but cannot claim to measure variability in the clinical sense. The thesis keeps that difference visible in its feature names and in its interface wording.

Heart rate also responds to walking, caffeine, illness and posture. For a caregiver who is on their feet through a shift, a raised heart rate alone says little about stress. That is part of the reason PULSE compares a person with their own expected heart rate, and not with a population norm.

2.2.4 Gradient-boosted trees for tabular data

Gradient boosting builds a sequence of shallow decision trees, each one fitted to the errors left by those before it. LightGBM is an implementation designed for speed and low memory use on tabular data [17]. Shwartz-Ziv and Armon compared deep models with tree ensembles on a set of tabular benchmarks and found that the ensembles were competitive, and that results depended on the data and not on a general rule [18]. The choice of LightGBM in this work follows from the shape of the data, which is a table of a few thousand rows and sixteen aggregate features per row, and from the experiments reported in Chapter 6. The choice does not imply that other model types could not perform better.

2.2.5 Evaluation when rows are not independent

A single caregiver contributes dozens of assessments, and those rows resemble one another. Saeb and colleagues showed that splitting such data by record, so that the same person appears in training and test sets, can produce estimates far more optimistic than the use case warrants, and that splitting by person approximates deployment on

new individuals [8]. Blagus and Lusa made a related point about resampling and cross-validation: operations that use label information, such as oversampling, must be applied inside the training folds if the assessment is to stay unbiased [9]. PULSE follows both lessons. Participants are assigned to groups before any feature is built, and every fitted component, including the imputer and the class weights, is fitted on training participants only.

The main metric is the area under the receiver operating characteristic curve. It measures how often a randomly chosen elevated case receives a higher score than a randomly chosen non-elevated case, and it does not depend on a single threshold [19]. It is not accuracy, and it says nothing about calibration. Both limits are repeated where results are reported.

2.2.6 Explainable predictions and the risk of misplaced trust

SHAP assigns each feature a contribution to a single prediction, built on Shapley values from cooperative game theory [10]. For tree ensembles the contributions can be computed exactly and efficiently, and they can be summarized over many cases to give a global picture [11]. Doshi-Velez and Kim argue that interpretability should be assessed against the task it is meant to support, not assumed [20]. A SHAP bar explains the model. It does not explain the person, and it does not show cause.

Risk scores can also cause harm when they stand in for something they do not measure. Obermeyer and colleagues found that a widely used health-management algorithm underestimated the needs of Black patients because it predicted healthcare cost and not illness [21]. The lesson for PULSE is to be exact about what the score represents, and to keep a person in the loop.

2.2.7 Design science and workforce allocation

Design science research treats the construction and evaluation of an artefact as a legitimate way to produce knowledge, provided the artefact addresses a real problem and is evaluated rigorously [22]. That description fits this project, in which the pipeline and the application were built and revised in turn. On the allocation side, Burke and colleagues surveyed nurse rostering, a field concerned with building shift schedules under many constraints [23]. Formal rostering optimizes a whole schedule. PULSE

does something much smaller, suggesting one substitute for one flagged caregiver under a handful of explicit rules.

2.3 Related Research

The studies closest to this work fall into three groups: datasets and methods for detecting stress, studies of healthcare staff, and work on interventions. Table 2.1 compares them with PULSE.

Several studies established that wearable signals can separate stress states. Healey and Picard did so for drivers [16], and WESAD provides a multimodal wearable dataset recorded under a laboratory protocol for stress and affect detection [24]. Gedam and Paul reviewed the use of wearable sensors and machine learning for mental stress detection [25]. These works show feasibility. Their populations and settings, however, differ from the ordinary working day of a caregiver, and most rely on labels collected under controlled conditions.

TILES-2018 moved closer to the real setting. It followed 212 hospital workers for about ten weeks, with wearable physiology, activity, proximity and repeated surveys [1]. It is the dataset behind this thesis. A review by Rana and colleagues of passive artificial-intelligence approaches to stress and burnout among frontline workers highlighted barriers to validation and generalization [26], and Cahoon and Garcia examined the generalizability of continuous stress monitoring for healthcare workers across real-world datasets [12]. Both point the same way. Good performance inside one dataset does not transfer automatically to another, and this thesis accepts that conclusion for itself.

Liu and colleagues proposed HiPAL, a deep framework that predicts physician burnout from the time-stamped activity logs of an electronic health record [27]. It shows that temporal occupational prediction is an active line of work. It differs from PULSE in population, input and outcome, and the thesis makes no claim that temporal prediction for workers is absent from the literature. On the intervention side, Cho and colleagues developed an AI-based tailored mobile intervention for nurse burnout and evaluated it in a single-arm trial [28]. It addresses what to do after burnout is recognized, which is a question PULSE leaves to supervisors.

Table 2.1: Closest related studies compared with PULSE

Study	Population or setting	Signals or inputs	Aim	Relation to PULSE
Healey and Picard [16]	Drivers in real-world driving	Physiological sensors	Detect stress level	Shows feasibility outside the laboratory; not a workplace caregiver setting
Schmidt et al. (WESAD) [24]	Participants in a laboratory protocol	Multimodal wearable signals	Dataset for stress and affect detection	Controlled-condition labels; not used in PULSE
Gedam and Paul [25]	Review of wearable studies	Wearable sensors and ML methods	Survey mental stress detection	Background on methods; no design choice in PULSE rests on it alone
Mundnich et al. (TILES-2018) [1]	212 hospital workers, about ten weeks	Wearables, proximity, daily surveys	Longitudinal dataset	The data source for PULSE
Rana et al. [26]	Review of frontline workers	Passive sensing and AI	Review of stress and burnout detection	Stresses validation and generalization barriers that PULSE acknowledges
Cahoon and Garcia [12]	Healthcare workers, several datasets	Wearable data	Test generalizability of continuous monitoring	Motivates the caution about transfer to care homes
Liu et al. (HiPAL) [27]	Physicians	Electronic health record activity logs	Predict burnout over time	Different population, input and outcome; shows temporal work exists
Cho et al. [28]	Nurses	Mobile app with AI tailoring	Tailored burnout intervention, single-arm trial	Intervention side; PULSE only supports a supervisor's decision

Placed side by side, the studies show a pattern. Detection work is plentiful, work on healthcare staff is growing, and the proposals that reach as far as an action for a supervisor are rare. None of these shows that nothing similar to PULSE exists. It shows only that the specific combination tested here, a participant-separated personalized pipeline, a matched proximity test, a comparison with the previous response, and a reviewable staffing suggestion, was not found in the sources I could verify.

2.4 Existing Systems, Products and Techniques

Supervisors today rely mostly on conversation, observation and occasional surveys. A caregiver may tell a supervisor that she / he is tired, or the supervisor may notice it. These routes are valuable and entirely dependent on attention and trust. They scale poorly to a large team and say nothing about the weeks in which nobody asked.

Technical approaches that could support the supervisor differ in what they demand and what they deliver. Table 2.2 compares those considered for PULSE and records the decision taken for each.

Table 2.2: Techniques considered for PULSE and the decision taken

Technique	Strength	Limitation in this setting	Decision in PULSE
Single population threshold on heart rate	Simple to explain	Ignores individual resting levels and activity	Not used as the model; compared against personalized features
Separate model for each caregiver	Fully personal	Each person has too few labelled surveys to train a stable model	Not used; one shared model fed with personal history
Shared gradient-boosted trees [17], [18]	Strong on tabular data, fast, works with SHAP [11]	Cannot learn long sequences directly	Chosen for both stages
Deep sequence models such as the temporal fusion transformer [29]	Multi-horizon forecasting with attention-based interpretability	Needs more data and tuning; harder to audit	Considered earlier, not trained in the final pipeline
Graph centrality from colleague proximity	Captures who works near whom	Coverage was limited, and co-location is not support	Tested in a matched experiment and excluded
Formal nurse rostering [23]	Optimizes whole schedules under constraints	Needs full roster data and organizational rules that PULSE lacks	Replaced by a small transparent rule reviewed by a person

2.5 Gap Synthesis and Conceptual Framework

The review supports the narrow gaps set out in Section 1.4. Studies of this kind are plentiful, but fewer of them show the timing of their inputs, test on people the model has never seen, report a simple comparator, test proximity against a matched baseline, and then connect an explanation to a reviewable staffing step. Each of the gaps is bounded. None of them says earlier researchers erred.

Figure 2.1 shows the conceptual framework that follows. Inputs available before a survey enter a first stage that learns an expected heart rate. The gap between the observed and expected rate enters the second-stage classifier. The classifier's output is explained with SHAP, accumulated into a sustained-risk rule, and passed to a rule that suggests a same-unit colleague. A supervisor reviews the suggestion, and that review is the end of the automated chain.

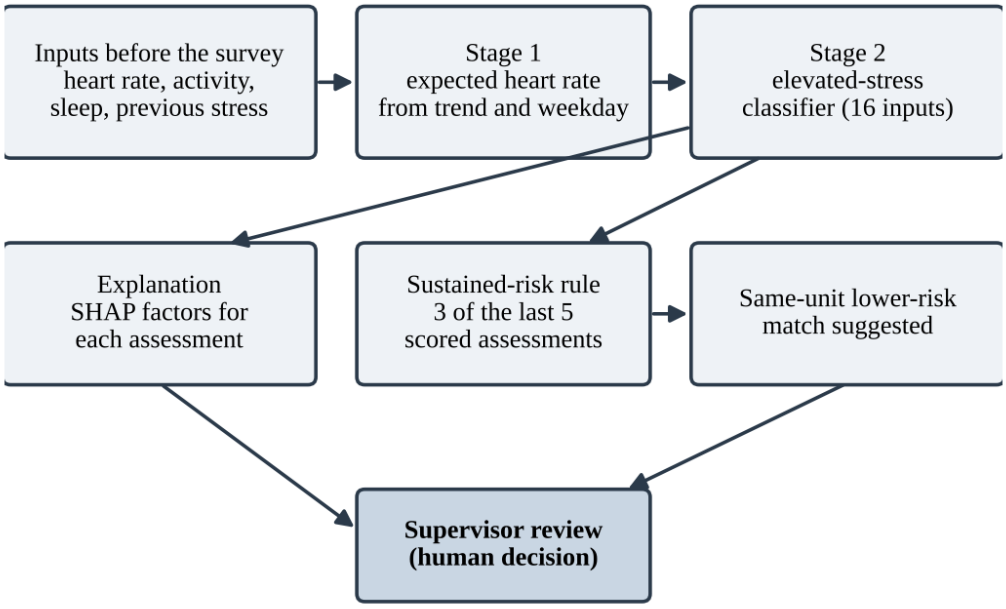


Figure 2.1: Conceptual framework of the PULSE pipeline

Table 2.3 links each gap to the response built into PULSE and to the place in the thesis where the evidence appears.

Table 2.3: Research gaps, the response in PULSE and where the evidence appears

Gap	Response in PULSE	Evidence
Unclear timing of inputs	Survey cut-off features, and a sensitivity analysis for the one exception	Section 3.3.1, Section 6.1.5
Transfer to new people	Participant-level allocation before features; grouped validation; caregiver bootstrap	Section 6.1.2, Section 6.1.3
Added value of proximity	Matched comparison across three model families	Section 6.1.6
Strength of the simple comparator	Paired comparison with the previous stress response alone	Section 6.1.4
Personalized representation	Shared model fed with personal trends; learned expected heart rate	Section 3.3.2
Explanation to action	SHAP factor lists, sustained-risk rule and reviewable match	Section 4.4, Section 6.2

3 RESEARCH METHODOLOGY

3.1 Research Design

This study uses a quantitative and experimental design, combined with the design science approach, in which a practical problem is identified, an artefact is built to address it, and the artefact is then evaluated [22]. The problem is the lack of a transparent way for supervisors to see how caregiver stress develops. The artefact is the PULSE module, which consists of a prediction pipeline and a web application. The two parts are evaluated in different ways. The prediction pipeline is evaluated through experiments on caregivers who were not used in training, and the application is evaluated against its requirements and demonstrated on historical data. Several datasets were examined before the final choice was made. These were the nurse stress dataset of Hosseini et al. [13] and the speech emotion datasets RAVDESS [14] and CREMA-D [15], which were explored during the first progress phase. TILES-2018 [1] was selected for the final research, because it combines continuous wearable recordings with repeated self-reported stress ratings from working hospital staff.

The experiments follow a fixed order. First, the participants of TILES-2018 were allocated to training, validation, test and demonstration groups before any feature was built. The allocation was made by person and not by record, using a fixed random seed. This means that all the records of one caregiver stay in one group, so the model is never tested on a person it has already seen. The allocation is described in Section 3.2. Second, features were built from the information available at the time of each survey. Third, a first-stage model learned the heart rate expected for each caregiver, and a second-stage model estimated the probability of elevated stress from the difference between the observed and expected heart rate and other features. Fourth, the final model was evaluated on the 27 caregivers in the test group. Four supporting analyses accompany this evaluation. A bootstrap that resamples caregivers measures the uncertainty of the result. A repeated participant-grouped validation is carried out on the development caregivers. A sensitivity analysis examines the one feature with a timing exception. A matched experiment tests whether proximity features add

information. Finally, a supplementary set of experiments, described in Section 3.3.5, examines whether the previous stress rating can be removed from the model.

3.2 Data, Participants and Study Materials

3.2.1 Source and provenance

The data come from TILES-2018, which was collected at USC Keck Hospital in Los Angeles by a research team at the University of Southern California, with Dr. Shrikanth Narayanan as the principal investigator [1]. The study followed 212 hospital workers for about ten weeks each. It recorded their physiological and activity signals through wearable devices, their Bluetooth proximity to colleagues, and their answers to repeated surveys.

I obtained access under a signed data use agreement with the University of Southern California. The agreement was signed on behalf of the university by Ms. Judy Genovese, Senior Transactions Officer at the USC Stevens Center for Innovation. Once the agreement was in place, the research team provided access to the data on their secure file transfer server and sent me the login credentials. I downloaded the files to my computer with the FileZilla client in order to inspect them. I also connected Google Colab directly to the same server, and I built the notebooks and trained the models in Google Colab. The downloaded files, the saved participant allocation and the results were stored in Google Drive.

Each directory of the data set contains a README file that describes its files, columns and units. I read these files before writing any code, and they were my main guide to the structure of the data. The care home visits described in Chapter 1 gave context for the requirements of the module, but they provided no data for the models.

Table 3.1: Source files used in the PULSE research pipeline

Source	Content	Use in PULSE
Heart-rate files, one per participant	Time-stamped photoplethysmography heart rate from a wrist tracker	Same-day summaries before each survey; historical daily means
Daily summary files	Steps, activity-zone minutes, resting heart rate and sleep efficiency	Previous-day activity features and sleep efficiency

Source	Content	Use in PULSE
EMA response file	14,795 survey rows with a stress rating from 1 to 5, and send, start and completion times	Outcome label, send time as prediction cut-off, previous-response feature
Bluetooth proximity files	Receiver detections with signal strength	Co-location network and centrality, used only in the matched experiment
Participant information	Anonymized identifiers	Joins and grouped splits only; not a model input

.

3.2.2 The survey and the outcome label

Each participant received one daily survey on job, health or personality topics. Day-shift workers received it at one of 6:00 a.m., 12:00 noon or 6:00 p.m., and night-shift workers at one of 6:00 p.m., midnight or 6:00 a.m. These are alternative send times and not three surveys a day. A separate second daily survey about psychological flexibility was sent at random times and carries no stress item [1]. The stress question reads “Overall, how would you rate your current level of stress?” and is answered on a five-point scale from “No stress at all” to “A great deal of stress”.

I converted the rating into a binary label. A response of 3 or above, meaning some stress or more, was labelled elevated, and a response of 1 or 2 was labelled not elevated. Responses with no rating were dropped. The cut-off is a choice made for this research. It is not a clinical threshold and it is not part of the original study. The original paper names its daily stress measure STRESSD, while the released field is called stress, and the pipeline used the released field only.

Two small inconsistencies in the source are noted and not resolved. The paper’s instructions describe a six-hour response window while its methods section gives a four-hour expiry, and the pipeline does not rely on either figure. Survey send time was used as the prediction cut-off because it is the conservative option. Scheduled dispatch is not the same as the moment of response, but using it keeps later readings out of the features.

3.2.3 Allocation of participants

Before any feature was built, every participant was assigned to exactly one of four groups with a fixed random seed of 42. The assignment was saved and is loaded, not

regenerated, by later runs. Table 3.2 shows the allocation. Training people fit the models, validation people choose when training stops, and test people are used only for final assessment. A fourth group of fifteen was set aside for the application demonstration and never entered any modelling.

Table 3.2: Participant allocation and usable rows

Group	People allocated	People with usable rows	Usable EMA rows	Rows after Stage 1 filtering	Purpose
Training	140	132	5,989	5,977	Fit models, imputer and class weight
Validation	30	30	1,296	1,292	Early stopping and model selection
Test	27	27	1,407	1,406	Final held-out assessment
Demonstration	15	not used	not used	not used	Application replay; excluded from development
Total	212	189	8,692	8,675	

Allocated people and usable people are different counts. Some allocated participants had missing files or produced no survey with usable heart-rate data before the cut-off, and the records do not give a complete ledger of every exclusion. The thesis therefore reports the counts that were observed and does not infer the rest.

3.3 Experimental and Study Procedure

Figure 3.1 gives the overall pipeline. Each step that follows is described in the order it runs.

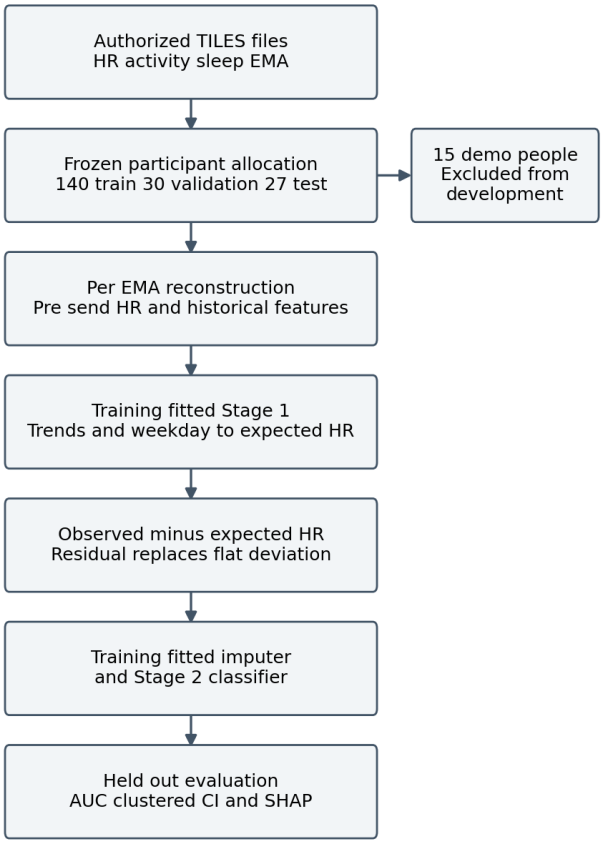


Figure 3.1: Research pipeline with participant separation and training-fitted stages

3.3.1 Reconstructing features at the survey cut-off

For every valid survey, sent at time T on date D, the pipeline selected heart-rate readings from date D with timestamps strictly before T. A survey with no such reading was skipped. From those readings it computed the mean, the sample standard deviation, the minimum and the maximum. These four values describe how the person's heart rate looked earlier that day.

History came from earlier dates only. The personal baseline was the mean of the daily mean heart rates on all days before D, giving each available day equal weight. For each of those days the pipeline subtracted the same baseline and averaged the last three and the last seven available deviations. These windows count available days, so gaps in the

record can stretch them over more than three or seven calendar days. A survey with no prior-day history was skipped.

Activity and resting heart rate came from the daily summary for date D minus one, left missing if absent. Sleep efficiency came from the summary for date D, which the pipeline treats as the previous night's completed sleep. The steps deviation was the previous day's step count minus the participant's mean over the whole study. That last feature is the one place where the main run used information from after the survey, because the study-wide mean includes later days. I disclose it here, rebuild the feature with earlier days only in Section 6.1.5, and do not call the main pipeline strictly causal.

The previous stress response was added by sorting each person's usable rows by survey time and shifting the raw 1 to 5 rating down by one. It is the previous usable response and not necessarily yesterday's, because a person can answer more than once a day or skip days. The first response of each person has no predecessor, which left 189 missing values. The pipeline orders responses by send time and does not check that the earlier response had been completed before the next survey was sent. That availability assumption is a limitation, and the thesis lists it as one in Chapter 7. Table 3.3 lists the sixteen inputs of the final classifier.

Table 3.3: The sixteen inputs of the final classifier

Feature	Construction and timing	Meaning and caution
hr_mean	Mean heart rate on the survey day before send time	Current level; not a stress diagnosis
hr_std	Sample standard deviation of those readings	Spread of sampled heart rate; not beat-to-beat variability
hr_min	Minimum of those readings	Lower end of the observed range
hr_max	Maximum of those readings	High excursions; exertion also raises it
hr_mean_deviation_model	Observed pre-send mean minus Stage 1 expected heart rate	Personal context given recent trend and weekday
hr_dev_roll3	Mean of the last 3 available prior-day deviations	Short historical trend
hr_dev_roll7	Mean of the last 7 available prior-day deviations	Longer trend; not necessarily seven consecutive days
NumberSteps	Step count of the previous day	Activity proxy; not a staffing workload
Cardio_minutes	Previous-day cardio-zone minutes	Activity intensity context
Fat Burn_minutes	Previous-day fat-burn-zone minutes	Activity intensity context, zones defined by the device

Feature	Construction and timing	Meaning and caution
Peak_minutes	Previous-day peak-zone minutes	High-intensity activity context
Out of Range_minutes	Previous-day minutes outside the device zones	Retained as supplied; no clinical meaning is assumed
RestingHeartRate	Resting heart rate in the previous-day summary	Resting physiological context
steps_deviation	Previous-day steps minus the participant's study-wide mean	Personal activity departure; timing caveat above
Sleep1Efficiency	Sleep efficiency in the summary for the survey date	Recovery-related proxy, not a measure of recovery time
lag1_stress	Raw rating of the previous usable response, 1 to 5	Persistence of reported stress; the dominant signal

The weekday used by Stage 1 is a separate input and is not a seventeenth classifier feature. Participant identity, the current stress rating, proximity centrality and the application's display names are excluded from the classifier.

3.3.2 Stage 1: a learned expected heart rate

Stage 1 is a LightGBM regressor with three inputs, the three-day and seven-day historical deviations and the day of the week, where Monday is 0. Its target is the survey-day mean heart rate before the cut-off. It is fitted on 5,977 training rows with complete inputs, using 100 trees, a depth limit of 3 and a learning rate of 0.05, and the seed 42. All other settings are library defaults.

The regressor is shared across people and never sees an identifier. It gains its personal character only from the history that enters its inputs. The residual,

$$r = HR_observed - HR_expected \tag{3.1}$$

is expressed in beats per minute, and it replaces the flat deviation in the classifier as *hr_mean_deviation_model*. It is a heart-rate residual and not a stress probability. The training residuals are in-sample, and I did not generate them out of fold, a point returned to in Chapter 7.

3.3.3 Stage 2: the elevated-stress classifier

Stage 2 is a LightGBM classifier on the sixteen features. Missing values are filled with medians computed from the training matrix alone, and the same fitted imputer

transforms the validation and test rows. The imbalance between classes is handled with a weight on the positive class,

$$w = N_negative / N_positive \tag{3.2}$$

computed from training labels only. In the final run the elevated share of training rows was 0.2398. Early stopping uses the validation AUC, never the test AUC, with a maximum of 500 rounds and a patience of 20. Table 3.4 lists the settings.

Table 3.4: Stage 2 classifier settings

Parameter	Value	Purpose and caveat
n_estimators	500	Upper bound; the final run stopped at 11 rounds
max_depth	3	Limits tree complexity
learning_rate	0.05	Smaller boosting steps
min_child_samples	30	Avoids leaves supported by very few rows
reg_alpha, reg_lambda	0.5, 0.5	L1 and L2 penalties
subsample	0.8	Configured fraction; no positive subsample frequency was set, so row sampling may not be active
colsample_bytree	0.8	Feature fraction per tree
scale_pos_weight	negatives divided by positives	Class imbalance adjustment from training labels
random_state	42	Reproducibility
Early stopping	20 rounds on validation AUC	Selected iteration 11 in the final run

The settings are those of the final run. A shallower, more heavily penalized configuration replaced an earlier one after the overfitting diagnosis described in Section 6.1.1.

3.3.4 Proximity network and the matched centrality experiment

The proximity files record which fixed receiver detected each participant's Bluetooth badge and how strongly. I floored each timestamp into five-minute windows and, for each participant and window, kept the receiver with the strongest signal. Within a date, window and receiver group that contained at least two people, every pair was joined by an edge, and edges were made symmetric. Degree centrality counted the distinct colleagues a person was co-located with on a day, and weighted centrality counted the number of co-location events. Degree here is a raw count and is not divided by the number of possible colleagues.

Of 197 files read, 190 were included and 7 failed. The 391,632 located rows contained 29,991 multi-person groups, giving 37,850 pairwise edges and 3,712 daily centrality rows. Joining these to the individual feature table left 2,989 rows with a network value, of which 2,018 were training, 500 validation and 471 test. Missing coverage was not filled with zero.

An earlier proxy, which counted the receivers that detected a person, measured movement and not contact, and I replaced it with true co-location. In the matched experiment, Model A used fourteen same-day physiology and activity features and Model B added the two centrality values. Both were trained and tested on identical rows with training-fitted median imputation. Three families were run: a class-balanced random forest, a class-weighted XGBoost model and a class-weighted LightGBM model. The XGBoost and LightGBM settings were 100 trees, depth 4, learning rate 0.1 and seed 42. This sixteen-feature Model B is a different object from the final sixteen-feature classifier, which uses no centrality.

3.3.5 Supplementary experiments without the previous stress response

After the main work, five further experiments, called V1 to V5, asked whether a questionnaire-independent model was feasible. V1 removed *lag1_stress* and compared fifteen-feature and sixteen-feature classifiers on identical test rows, using a feature rebuilt with strictly earlier step history. V2 tuned twenty-four configurations from four model families on validation data. V3 tried support vector machines, neural networks, extra-trees and histogram gradient boosting on the original fifteen features and on

twenty-six features with eleven engineered combinations. V4 was a descriptive audit of raw heart-rate samples in short windows before each survey, restricted to training people, and trained no classifier. V5 tested seventeen new short-window and personal-baseline features in a controlled comparison on validation data. After V1, no experiment touched the test cohort or the demonstration group.

3.4 Measures and Evaluation Metrics

Table 3.5 summarizes the measures and what each one can and cannot say.

Table 3.5: Evaluation measures and their limits

Measure	Purpose	Limit
ROC AUC [19]	Ranks elevated against not-elevated cases across all thresholds; main discrimination metric under imbalance	Not percentage accuracy and not calibration
Accuracy at 0.5	Share classified correctly at the model's default threshold	The majority baseline is already 78.73% on the test group
Caregiver-bootstrap interval	Resamples people while keeping each person's rows together	Describes uncertainty within this cohort only
Repeated grouped cross-validation	Stability over held-out development people	Fold results are related; mean and SD are not a confidence interval
Paired bootstrap difference	Compares two models on identical rows	Depends on the row population chosen
Mean absolute SHAP [11]	Global size of each feature's contribution	Not a percentage and not a cause
Precision, recall, F1, PR AUC, Brier score	Used in the supplementary experiments where saved	Valid only with the threshold and cohort stated

The main measure is the area under the receiver operating characteristic curve, called ROC AUC [19]. It describes how well the model ranks cases. If one assessment with elevated stress and one without are chosen at random, the AUC is the probability that the model gives the higher estimate to the elevated one. A value of 0.5 means no better than chance, and 1.0 means a perfect ranking. I chose it because elevated stress is the less common outcome, and because it does not depend on a single threshold. It does not show how accurate the probabilities are, and it does not give the share of cases that are classified correctly.

Accuracy at the default threshold of 0.5 is reported for context. On the test group, a rule that always predicts “not elevated” would already be correct for 78.73% of the rows, so accuracy alone says little about the model. Precision is the share of predicted elevated cases that were truly elevated, and recall is the share of truly elevated cases that were found. F1 combines the two. These measures depend on the threshold and on how common the outcome is, and they are reported only where the threshold and the group of people are stated.

Because the groups are small, the uncertainty of each result is also reported. The caregiver bootstrap gives a range for the AUC, by repeating the calculation on resampled sets of people. The repeated grouped validation shows how stable the result is when different caregivers are held out. The paired bootstrap difference compares two models on exactly the same rows. Finally, the mean absolute SHAP value shows how much each feature contributes to the model’s output [11]. It is a measure of influence and not of cause.

3.5 Data Analysis

The data analysis started before any model was trained. The TILES-2018 files had to be retrieved, checked and understood first, and the modelling table was built only after that. This section follows the order in which I did the work, and ends with the statistical analyses used to judge the results.

I worked in Google Colab, with the raw data, the cached tables and the results stored in a project folder on Google Drive. The files were downloaded from the provider’s secure FTP server over an encrypted connection. The password was typed in at run time and was never saved in a notebook. The download function skipped any file that was already present, so an interrupted session could continue without fetching everything again. Before opening the files in each data directory, I read the README that came with them. These notes explained the field names, the sampling interval of each sensor and how participants were identified. I then retrieved the full ecological momentary assessment (EMA) response file, which held 14,795 rows and 150 columns and covered all 212 participants. That check is the reason a file audit stayed in the workflow.

The stress question was the field I examined first, because the whole study depends on it. Of the 14,795 survey rows, 3,445 (23.28%) had no stress answer. Among the 11,350 rows that did, 5,702 were rated 1 (no stress at all), 2,814 were rated 2 (very little), 2,270 were rated 3 (some), 433 were rated 4 (a lot) and 131 were rated 5 (a great deal). Ratings of 3 or more therefore made up roughly a quarter of the answered rows, which makes elevated stress the minority outcome. This fact later explained why accuracy was a weak measure and why I weighted the classes in training. I also parsed every timestamp and found none that could not be read. The surveys ran from 5 March 2018 to 13 July 2018. I counted the rows by survey type and by each participant's work field, to see which parts of the hospital the data came from, and saved the checked table as a parquet file so that later notebooks started from the same verified copy.

Next I made an inventory of the sensor files. Fitbit heart-rate, daily-summary, sleep and step files existed for 209 participants, while the OMsignal garment data, the Owl location data and the Jelly proximity data covered smaller groups. The OMsignal files had a high share of missing values, and the Fitbit signals had the widest coverage. A model meant for ordinary working conditions needed the signal that most people actually had, so the Fitbit streams became the main wearable input. I tested the preparation steps on a few participants before applying them to everyone. For each person and each day, I calculated the mean, standard deviation, minimum, maximum and number of heart-rate readings, and aligned these with the survey responses by date. Step counts were handled the same way. When this worked on the sample, I ran it on the whole cohort and saved the daily feature table. The final pipeline later replaced the daily summaries with a window of time before each survey, so that every input could be tied to a moment earlier than the answer it was meant to predict. The timing rule is described in Section 3.3. Several steps in the audit notebook changed when I moved to separate notebooks, so the figures in the rest of this section come from the final consolidated notebook. The demonstration group was downloaded separately and kept out of all modelling. Of the 212 participants allocated, 197 formed the modelling pool and 189 had usable rows. Four heart-rate files could not be read and were dropped. The first survey of each person had no earlier answer to serve as the previous-response feature, so that row was removed, which affected 189 first responses. The usable rows

were 5,989 for training, 1,296 for validation and 1,407 for testing, giving 8,692 in total. After the filtering needed for the first-stage heart-rate model, 5,977, 1,292 and 1,406 rows remained, a total of 8,675.

The proximity data needed a separate step. The Jelly sensors record when two participants are close to each other. I grouped these records into five-minute intervals, built a co-location network and calculated centrality measures for each person. Of the 197 modelling participants, 190 Jelly files were read and 7 failed. To find out whether the network added anything, I compared models with and without the centrality features on the same 2,989 matched rows.

3.6 Ethics, Privacy, Security and Responsible AI

The research uses an existing dataset of de-identified records, so no new participants were exposed to risk and no new data were collected from individuals. The data use agreement sets the working rules, and the thesis follows them. The data are used only for internal research tied to the project. The provider's FTP credentials are kept private. No attempt was made to identify participants, to obtain a coding key or to contact them. Third parties were given no access to the data without prior written consent from the university that owns it. The data contribution is acknowledged, aggregate results may be reported, and participant-level data are not published. Unauthorized use or disclosure must be reported within fifteen days of awareness, and the agreement carries retention, return and termination terms.

The application shows synthetic caregiver names and ward labels in place of anything real. Real shift and unit fields, which are anonymized, are used separately for matching.

Access control follows three roles and is built on Supabase. The server validates each bearer token with Supabase Auth, loads the person's role from their profile in the Supabase PostgreSQL database, and checks that the requested caregiver is in scope before it answers. Administrators see all monitored caregivers, and supervisors see their own team. The Supabase service-role key lives on the server only. Row-level security is enabled on the sensitive tables, but the service role bypasses it, which makes the checks inside each route essential and not optional. Candidate search for a

substitute deliberately crosses supervisor boundaries inside the same real unit, and the response is limited to the minimum information the review needs.

Four responsible-AI choices shaped the design. The score is described as a model estimate of elevated self-reported stress and never as a diagnosis. SHAP factors are presented as what moved the model, not as causes. The recommendation is a suggestion and a person decides, and marking it reviewed does not reassign anyone. And no automated step touches employment decisions. These choices respond to the warning of Obermeyer and colleagues that a risk score can drift away from the thing it is meant to represent [21]. Workplace monitoring carries its own sensitivity, and Sri Lanka's Personal Data Protection Act would be relevant to any real deployment that processes caregivers' personal data [30]. Reaching that stage would need informed consent, a lawful basis and an institutional ethics review that this prototype does not claim to have.

3.7 Reproducibility and Research Artefacts

All random operations use the seed 42, and every later run reads the saved participant allocation and does not create it again. The notebook reports the software versions it used: pandas 2.2.3, NumPy 2.1.3, scikit-learn 1.6.1, LightGBM 4.6.0, XGBoost 3.4.1, SHAP 0.52.0, SciPy 1.16.3 and statsmodels 0.15.0. The final notebook writes its outputs to a separate research folder, so the research runs cannot overwrite the files that the application reads. These outputs are the first-stage model, the classifier and the missing-value handler as three files, the SHAP plot, the twenty-five results of the repeated validation, the matched-model comparisons with their ranges, and the files of the sensitivity analysis. The code is maintained in a GitHub repository (https://github.com/dayanakumar-IT/R26-DS-010-Intelligent-Care-Support.git), which contains the notebooks and the application source.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements came from three sources. Consultations with the care home in April 2026 established the practical needs: regular insight into how caregivers are coping, and a route to substitute cover when one of them is struggling. The external supervisor added a caregiver's view of which features would actually be used. The data use agreement and the research findings set the constraints. Table 4.1 and Table 4.2 list the requirements, each with an identifier that the test cases in Chapter 5 refer to.

Table 4.1: Functional requirements

ID	Requirement	Origin
FR1	Authenticate each user and apply the role of administrator or supervisor	Access rules; data use agreement
FR2	List the monitored caregivers within the user's scope	Supervisor need for team overview
FR3	Show a caregiver's history of features, baseline and device records	Supervisor need for context
FR4	Estimate the probability and class of elevated stress for a chosen date, with contributing factors	Research objective 5
FR5	Show stored risk history, keeping unscored days visibly separate from low-risk days	Trend review; misreading risk
FR6	Show expected against actual heart rate for a caregiver	Research objective 2
FR7	Summaries risk across the team as counts, trends and a heatmap	Supervisor need for overview
FR8	Display metadata of the active model from a registry	Traceability
FR9	Flag sustained risk and propose a lower-risk colleague from the same real unit	Substitute-cover need (7 April interview)
FR10	List recommendations and record a human review without reassigning anyone	Responsible-AI principle
FR11	Allow an administrator to upload heart-rate, daily-summary and survey files to a staging area	Demonstration on historical data
FR12	Reveal historical days in order and score once enough history exists	Replay of historical readings
FR13	Return the reasoning when no eligible match exists, and do not guess	Transparency

Table 4.2: Non-functional requirements

ID	Requirement	Origin
NFR1	Security: validate tokens on the server, check scope on every route, and keep the service key off the client	Data use agreement
NFR2	Privacy: use anonymized identifiers and synthetic display names; return the minimum candidate information	Data use agreement
NFR3	Transparency: present factor directions, avoid diagnostic wording, state that scores are model estimates	Research findings; Obermeyer et al. [21]
NFR4	Reliability: tolerate missing or non-finite values and return clear error codes	Operational need
NFR5	Efficiency: load model artifacts once at start-up and reuse them	Operational need
NFR6	Usability: role-appropriate views, colour tiers that are not read as disease severity	Supervisor need
NFR7	Traceability: record model versions and review actions	Responsible-AI principle
NFR8	Integration: use the CareSense shared authentication and navigation	Group architecture
NFR9	Reproducibility: pair every model artifact with its imputer and feature order	Research method

4.2 Component Architecture within the Complete Solution

CareSense is a web application made up of four modules, and PULSE is one of them. The architecture is described at two levels. At the system level, the CareSense front end hosts all four modules behind a shared sidebar, login and navigation. PULSE contributes its own views, its own backend routes and its own tables, which are stored in the shared Supabase PostgreSQL database and protected by Supabase authentication and row-level security.

Figure 4.1 shows the PULSE service and the interfaces through which it communicates. The browser views talk to a FastAPI service. The service checks every request against the authentication provider, loads the model artifacts in its own process, and reads and writes the relational database. Historical research files reach the system only through the administrator's upload route, and the other CareSense modules reach PULSE through the shared front-end shell.

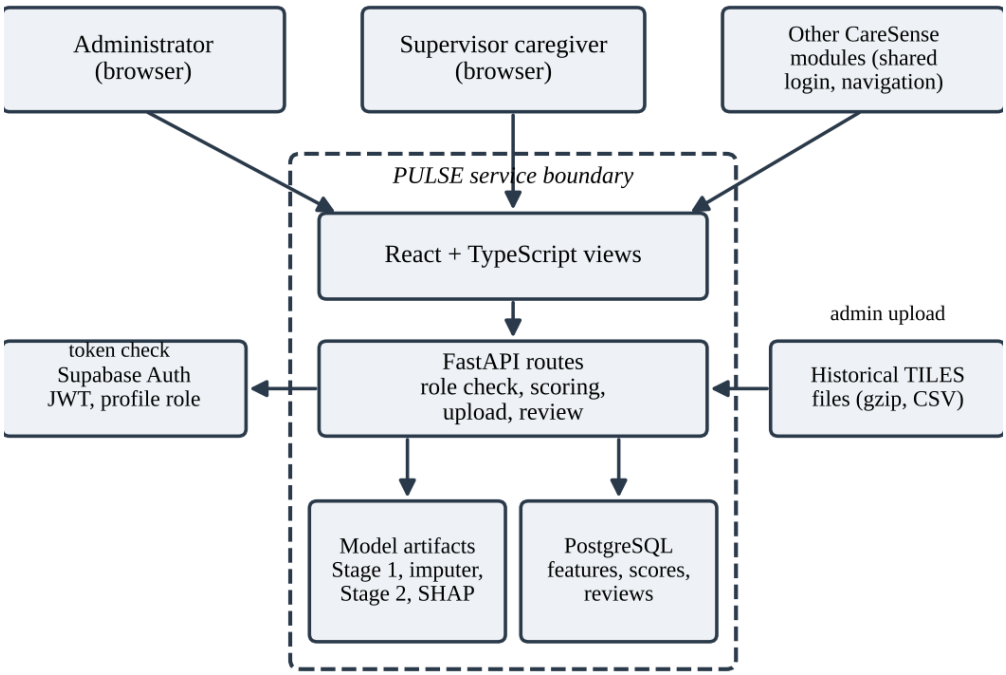


Figure 4.1: PULSE service boundary and external interfaces

Table 4.3 specifies each interface, what crosses it and the protection applied.

Table 4.3: Interfaces of the PULSE service

Interface	Between	Carries	Protection
Browser to views	Administrator or supervisor and React application	Page requests and user actions	Authenticated session in the shared shell
Views to service	React and FastAPI	REST requests with JSON responses	Bearer token on every call; cross-origin rules limited to the development origin
Service to authentication	FastAPI and Supabase Auth	Token validation and profile role	Server-side only
Service to database	FastAPI and PostgreSQL through Supabase	Features, scores, staging rows, recommendations	Service-role key held on the server; row-level security on sensitive tables
Service to models	FastAPI and joblib artifacts	Stage 1, imputer, Stage 2 and explainer in memory	Files placed in a models folder, loaded at start-up
Administrator upload	Administrator and FastAPI	Compressed heart-rate and summary files; survey CSV	Administrator role only; column checks; staging not production tables
Shared shell	Other CareSense modules and PULSE	Login, sidebar, navigation, audit and registry definitions	Shared authentication; module-specific tables

Interface	Between	Carries	Protection
Batch scoring	Administrator script and FastAPI	Historical dates sent to the scoring route	Administrator token supplied at run time, never stored in the script

4.3 Design Alternatives and Decisions

The decisions in Table 4.4 fall into four groups: the label and features, the proximity network, the modelling approach, and the application.

The first group concerns what the model learns from. The stress label is binary, because in the first phase the middle class of a three-class label made up only 6.6% of windows and was almost never predicted correctly. The features are rebuilt at the survey cut-off, and not taken from whole days, so that every reading used was recorded before the survey it predicts. The second group concerns proximity. The network was rebuilt as a true co-location network, because counting receivers measured movement and coverage and not contact with colleagues. Matched comparisons then showed no detectable gain from centrality, so it was left out of the classifier.

The third group covers the model. A single shared model uses each person's own history, since individuals have too few labelled surveys for a stable model of their own. Class weighting and regularization were added after the unweighted model favored the majority class and the unregularized lag model fitted the training data far better than the test data. A learned expected heart rate replaced a flat personal mean, and gradient-boosted trees were preferred over deep sequence models, because the data are a modest table of aggregate features. The last group covers the application. Redistribution uses a transparent rule instead of optimized rostering, because no roster, skill-mix or capacity data are available and a person must be able to read the reasoning. The data are replayed from historical days, since no live care-home sensor feed exists. FastAPI serves the models in the same process, and Supabase supplies authentication, storage and row-level security as one service.

Several design decisions in this project were revised after an early approach proved unsuitable during development. Table 4.4 records the main decisions, the alternative considered for each, and the reason it was set aside.

Table 4.4: Design alternatives and the decisions taken

Decision	Alternative set aside	Reason
Binary elevated-stress label	Three-class label	In the first phase the middle class made up only 6.6% of windows and was almost never predicted correctly
Features rebuilt at the survey cut-off	Whole-day features with whole-period baselines	Whole-day data cannot guarantee that every reading existed before the survey
True co-location network	Counting the receivers that detected a person	Receiver counts measured movement and coverage and not contact with colleagues
Proximity centrality left out of the classifier	Including it	Matched comparisons showed no detectable gain, with intervals that crossed zero
Class weighting and regularization	Unweighted model; unregularized classifier	The unweighted forest reportedly predicted only the majority class, and the unregularized lag model fitted training data far better than test data
Learned expected heart rate	Flat personal mean	A shared regressor fed with recent trends and weekday captured the individual context better in the observed comparison
One shared model with personal history	A separate model per caregiver	Each person has too few labelled surveys for a stable individual model
Gradient-boosted trees [17], [18]	Deep sequence models [29]	A modest table of aggregate features suits trees; earlier sequence-model plans were not carried through
Transparent rule for redistribution	Optimized rostering [23]	No roster, skill-mix or capacity data are available, and a person must be able to read the reasoning
Replay of historical days	Live wearable feed	The data are historical, and no live care-home sensor feed exists
FastAPI beside the models	A backend in another language	The artifacts are Python objects that load directly in the same process
Supabase for authentication and storage	Separate services	One service supplies tokens, roles, PostgreSQL and row-level security together

4.4 Detailed Design

4.4.1 Scoring design

A scoring request identifies a caregiver and a date. On receiving it, the service validates the access token, retrieves the caller's role and confirms that the requested caregiver falls within the caller's scope. It then reads the features stored for that day. The weekday is derived from the date, and together with the other two Stage 1 inputs it is passed to the regressor, which returns the expected heart rate. The residual is the stored heart rate minus this expected value. Sixteen values are then placed in a fixed order

for the classifier. Missing values are imputed by position and not by name, because the application uses lower-case feature names while the fitted objects retain the original capitalization. The classifier returns the probability of the positive class together with a label, and the explainer returns a contribution for every feature. The response contains the probability, the predicted class, all factors ranked by absolute contribution, the leading factor, the raw features, the expected heart rate and the residual.

For display, the probability is mapped to three tiers: 0.5 or above is high, 0.35 to below 0.5 is moderate, and below 0.35 is low. The cut-offs were set after inspecting an earlier distribution of scores. They are conventions for presentation and have not been validated clinically.

4.4.2 Data model

The data model is stored in Supabase's PostgreSQL database and is organized around two ideas. The first is that research data and application data are kept apart, and the second is that real identities never appear in the application. Table 4.5 lists the PULSE tables. The definitions come from forty-three migration files that also contain the other modules' tables, and only the PULSE and shared definitions are described here.

Table 4.5: Principal database tables used by PULSE

Table	Purpose
profiles	Authenticated user and role; referenced by supervisor and reviewer fields
participants	Anonymized TILES metadata and split group; parent of the research-derived tables
daily_features	One row per participant and date, with feature values, a pipeline version and later score columns
participant_baselines	Fixed historical or running baseline, with provenance dates
centrality_daily	Stored degree and weighted centrality, kept as historical data though excluded from the classifier
caregiver_profiles	Synthetic display identity linked one-to-one to a participant, with supervisor, synthetic ward, and real shift and unit
device_registrations	Display device metadata; a replay device label does not prove a connected wearable
model_registry	Component, model name and version, training time, metrics and artifact paths
audit_log	Shared append-only event definition

Table	Purpose
redistribution_recommendations	Flagged and suggested caregivers, recorded risk, reasoning, status and review metadata
raw_upload_staging	Imported daily values, attached stress rating and reveal state

At the center of the model is the participants table. It holds the anonymized TILES metadata, including the group to which each person was allocated, and it is the parent of the tables derived from the research data. Beneath it, daily_features stores one row per participant and date. The participant_baselines table keeps a fixed historical or running baseline for each person, with the dates that show where it came from. The centrality_daily table keeps the degree and weighted centrality values.

The application layer is separate. The profiles table links each authenticated user to a role, and the supervisor and reviewer fields in other tables refer to it. This is how the role checks in Section 3.6 find out who may see which caregiver. The caregiver_profiles table gives each participant a synthetic display identity, with a one-to-one link to the participant, a supervisor, and a synthetic ward, and with the real shift and unit retained. The names on screen are therefore not the names of TILES participants, while the unit stays real so that the redistribution rule has a meaningful place to search. The device_registrations table holds display information for devices.

Three tables support operation and accountability. The model_registry records each component with its model name, version, training time, metrics and artifact paths, so the application can state which model produced a score. The audit_log is a shared append-only event record. The raw_upload_staging table holds the imported daily values with the attached stress rating and its reveal state. Sensitive tables have row-level security enabled and no read policy for ordinary authenticated users, so reads go through the service, which applies the role checks described earlier. One item remains unreconciled. The backend reads and writes three score columns on the feature table that I could not find in the supplied migration files.

4.4.3 Interface design

The service exposes eighteen routes, which Table 4.6 groups by purpose. The interface follows REST conventions. Resources have their own paths, requests and responses

are JSON, and the status code tells the client what happened. Every request carries the user's access token, and no route is left open apart from the health check. Each view in the application reads from one group of routes, and the service returns data that is ready to show. A response includes the probability, the ranked factors and the expected and actual heart rate together, so the browser does not repeat any calculation. This keeps the model logic in one place on the server. Reading and scoring are available to any user in scope, the redistribution routes record who acted, and the replay controls are limited to administrators.

Two design choices deserve attention. First, scoring a date and storing a score are separate actions. The scoring route returns the full result for a caregiver and date without writing anything, so a user can examine a result without changing the record that later views and the redistribution rule rely on. Second, history is presented in two ways. Stored scores are read back as they were calculated, while the expected heart rate against the actual value is recomputed on request. This lets the interface show how a caregiver's pattern changes over time without keeping a second copy of the derived values.

Error handling is uniform: 403 for a role or scope denial, 404 for a missing caregiver or day or an exhausted replay, 400 for malformed uploads, and 500 for database or inference failures. A request outside a user's scope is refused with 403 and is never answered with an empty list. This avoids showing a supervisor a blank screen that looks like "no stress" when the real cause is a missing permission.

Table 4.6: Route groups of the PULSE service

Group	Routes	Purpose
Service	GET /health, GET /me	Health check; authenticated user and role
Caregivers	GET /caregivers; GET /caregivers/{id}/history, /risk-history, /baseline-history	Scoped list; features and devices; stored scores; recomputed expected against actual heart rate
Scoring	POST /caregivers/{id}/simulate	Stage 1, Stage 2 and explanation for a date; does not store the score
Analytics	GET /analytics/risk-summary, /risk-heatmap, /team-trends, /model-performance	Counts, heatmap, trends and registry metadata
Redistribution	POST /analytics/generate-redistribution-recommendations; GET /analytics/redistribution-recommendations; POST /analytics/redistribution-recommendations/{id}/mark-reviewed	Flag and match; list; record review

Group	Routes	Purpose
Administration	POST /admin/upload-raw-data/{id}, /admin/upload-survey-responses/{id}, /admin/reveal-next-day/{id}; GET /admin/staging-status/{id}	Stage uploads, attach surveys, reveal and score a day, report staging state

4.4.4 Redistribution design

Figure 4.2 gives the rule. Only historical caregivers are considered. Administrators evaluate everyone, and supervisors flag their own people, though the search for a substitute can reach other supervisors' caregivers in the same real unit. A caregiver needs at least five scored assessments, and is flagged when three or more of the last five were predicted elevated. These are scored assessments, not necessarily five consecutive days.

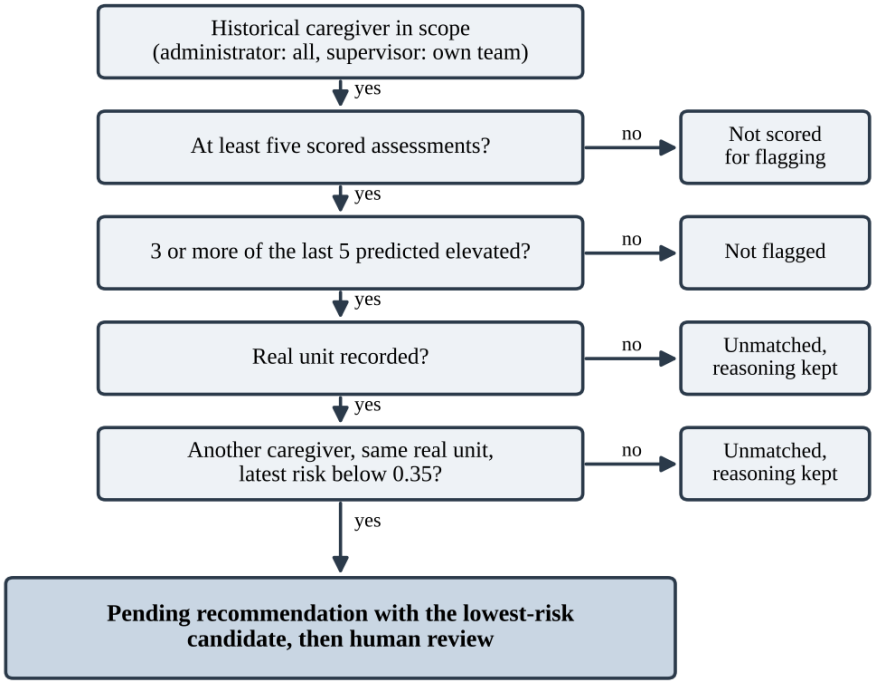


Figure 4.2: Decision flow of the redistribution rule

A candidate must be a different caregiver in the same recorded real unit, with a latest probability below 0.35. The lowest probability wins. If the flagged caregiver is on the day shift, or the candidate has a lighter recent activity load, the explanatory text says so, but neither fact changes the ranking. Activity load adds cardio, peak and fat-burn minutes, treating missing values as zero for this rule. If no real unit is recorded or no

candidate qualifies, the recommendation is stored as unmatched with its reasoning, and the synthetic ward labels are never used as a fallback. A caregiver already holding a pending recommendation is not flagged again, whereas a reviewed or dismissed one does not block a fresh recommendation.

4.4.5 Replay design

Because the data are historical, a demonstration needs a way to present them one day at a time. An administrator uploads compressed heart-rate and daily-summary files, which the service aggregates to one row per date and holds in staging, and may then attach the survey ratings. Each call to the reveal route promotes the earliest unrevealed date into the feature table. Earlier revealed days supply the running baseline and the three-day and seven-day deviations, and the previous revealed day's raw rating supplies the lag. The first seven revealed days are not scored. Scoring begins on the eighth, once seven earlier days exist.

Replay differs from the research design in three ways that the thesis states so nobody misreads a demonstration as a reproduction. Replay uses full-day heart rate and current-day activity, where the research used pre-survey readings and previous-day activity. Replay computes the steps deviation from earlier revealed days including the current one, which differs from both the main run and the strictly prior sensitivity run. And the reveal writes to the feature table and then updates staging as two separate operations without a shared transaction.

4.5 Standards and Professional Practice

The work adopts the practices that applied to each part of it. Interfaces follow REST conventions, with resource-style paths, JSON bodies and meaningful status codes. Authentication uses signed tokens, and authorization is checked server-side on every route. The database applies least privilege through row-level security and explicit grants, and the service role is confined to the server environment. Schema changes are kept as ordered migrations, and models are recorded in a registry with their versions.

On the research side, participants are separated before features are built, every fitted component is fitted on training people only, and the numbers reported carry intervals

computed over people. Negative and exploratory results are labelled as such. Sources are cited in IEEE style and checked against primary records. On the ethical side, the thesis follows the data use agreement, keeps a person in the loop, and avoids diagnostic claims. No external standard for medical devices or clinical decision support is claimed, because PULSE is a research prototype.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The research and the application were built in separate environments. The research pipeline ran in Google Colab, using Python notebooks. The TILES-2018 files were retrieved from the provider's secure FTP server, inspected with FileZilla, and stored in Google Drive together with the cached tables and the saved results. Data preparation and statistics used pandas, NumPy, SciPy and statsmodels, and the models were built with scikit-learn, LightGBM and XGBoost, with SHAP for the explanations. The application was developed locally. Its backend is a FastAPI service run by Uvicorn, which loads the saved model files with joblib and reads and writes data through the Supabase Python client. Its front end is written in React with TypeScript and CSS modules, and it runs as the PULSE views inside the CareSense shell. Supabase supplies the PostgreSQL database and the authentication, with roles, access tokens and row-level security. The source code for both parts is maintained in a GitHub repository. Table 5.1 lists the main technologies, the versions where they were recorded, and the role of each.

Table 5.1: Technology stack

Layer	Technology	Role
Research	Python with pandas 2.2.3, NumPy 2.1.3, SciPy 1.16.3, statsmodels 0.15.0	Data preparation, statistics and bootstrap
Research	scikit-learn 1.6.1, LightGBM 4.6.0, XGBoost 3.4.1	Imputation, models and the matched experiments
Research	SHAP 0.52.0	Tree explanations
Backend	FastAPI served by Uvicorn	Authorized routes for scoring, history, analytics, upload and redistribution
Backend	pandas, NumPy, joblib, SHAP, httpx, Supabase Python client, dotenv	Feature preparation, artifact loading, explanations, database access and configuration
Frontend	React with TypeScript and CSS modules	PULSE views inside the CareSense shell
Data and identity	Supabase PostgreSQL and authentication	Roles, tokens, relational storage and row-level security

5.2 Component Implementation

5.2.1 Research pipeline

The research code is organized as one consolidated notebook that rebuilds the main analysis from the raw files, one section at a time. It loads the frozen participant allocation, reconstructs the features at the survey cut-off, adds the previous response, trains the two stages and evaluates them on the held-out people. The bootstrap, the repeated validation, the sensitivity analysis, the matched centrality experiments and the explanations follow. Nothing in it can overwrite the files that the application reads. Its outputs go to a separate folder, and its three model files carry names that differ from those the application loads, so an update needs a documented mapping and not a file copy.

Several details of the implementation affect correctness and are easy to get wrong. The survey send time is parsed with invalid values coerced to missing and then dropped. Same-day heart-rate readings are selected with a strict less-than comparison against that time. The previous response is created with a grouped shift, after sorting by person and time. Imputation and the class weight are calculated from the training matrix and applied unchanged to the other groups. The bootstrap keeps all rows of a person together when it resamples, so one person can appear several times in a draw.

A second line of work asked whether the module could predict without any previous survey response, which would let it run without asking caregivers anything. I carried this out as five supplementary experiments, V1 to V5, each in its own notebook and its own output folder, kept apart from the main run. V1 removed the previous response from the classifier and compared the result with the full model on the same 1,406 test assessments from 27 caregivers. The test AUC fell from 0.7213 to 0.6108. It also rebuilt the steps deviation from strictly earlier days and checked which features could be supplied at the time of a request. V2 tried 24 configurations of LightGBM, XGBoost, CatBoost and logistic regression on the training and validation groups only, and the best validation AUC was 0.5623. The threshold that maximized F1 flagged 96.5% of assessments as elevated, which is not usable as an alert. V3 tested 18 candidates, including support-vector machines, neural networks and extra trees, on the

original 15 features and on 26 features with 11 engineered ones. The best AUC was 0.5661, from a neural network on the original features, so the added features did not help. V4 trained no classifier. It examined the raw heart-rate readings of the training caregivers, whose median sampling interval was 5 seconds, and checked how many readings fell in windows of 15 to 120 minutes before a survey. V5 returned to the two-stage design with 17 new short-window heart-rate features, and the validation AUC was 0.5572 against 0.5360 for the baseline set. None of the five experiments produced a useful replacement for the previous response. I therefore kept the main model as the result of this thesis and treat V1 to V5 as supplementary, negative findings. The results of V2, V3 and V5 come from the validation group and those of V1 from the test group, so their AUC values are not directly comparable.

5.2.2 Backend service

At start-up the service loads the Stage 1 regressor, the Stage 2 classifier and the imputer once, builds the explainer and checks that the classifier has a positive class. A missing model file stops start-up with an error, which is the behavior wanted, since a service that answers without its model would be worse than one that does not run. Figure 5.1 shows how a scoring request moves through the service.

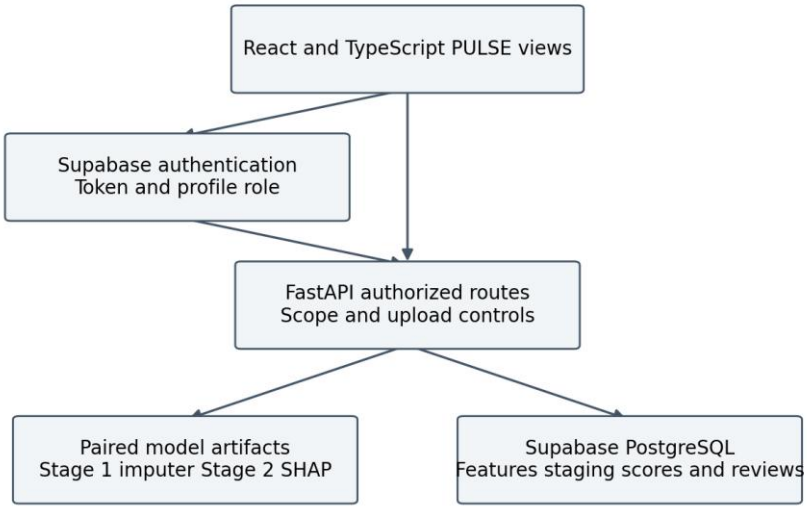


Figure 5.1: PULSE service architecture and authorization boundaries

Source: Author's own work.

Several choices in the scoring code come from problems met along the way. The application's feature names are lower case, whereas the fitted imputer was trained with the original names, so imputation is done by position and the order is fixed in one constant. The index of the positive class is looked up and not assumed. The explainer can return its values in more than one shape, and the code handles the supported ones. Missing or non-finite values become JSON nulls in the response so that the front end can tell an unavailable value from a zero. When the database connection drops after sitting idle, the client recycles idle connections after thirty seconds and retries a failed request once. The comments in the code justify the retry as safe in general, a claim I did not prove, so I treat it as a pragmatic fix for a real fault and not as a guarantee.

Two behaviors of the scoring and history routes are worth recording. The scoring route computes a result and returns it without storing it. The baseline history route recomputes Stage 1 for every scored row, because the expected heart rate is not stored.

5.2.3 Frontend views

The PULSE module has six tabs: Overview, Risk Analysis, Caregiver Profiles, Trends, Redistribution and Demo Control. The overview shows counts of caregivers in each tier, an attention list and a team heatmap. These counts describe the caregivers visible in the application and not the 212 people of the research dataset. Risk Analysis fetches one assessment and shows the probability, the factor lists and a history chart with a heart-rate overlay. The profile tab has sub-views for overview, expected and actual baseline, risk history, workload and explanations. The trends tab summarizes stored histories. The redistribution tab generates and lists recommendations. Demo Control holds the upload and replay tools for administrators.

The factor list shows up to three factors that raise the estimate and up to three that lower it. The backend returns every factor sorted by magnitude, and the front end translates selected names into plain words. A contribution of exactly zero is classed as reducing, because the backend uses a greater-than-zero test, so the absence of a positive contribution must not be described as proof of protection. The heart-rate spread feature is labelled in the interface with wording that suggests heart rate variability. It is a sampled spread and not true variability, and the label should be

corrected. The avatar keeps one fixed shape, and its color indicates the display tier only.

5.2.4 Administrator upload and replay

The upload route accepts compressed heart-rate files with Timestamp and HeartRatePPG columns, and compressed daily-summary files with the activity, resting heart rate and sleep efficiency columns. It aggregates heart rate by date into mean, standard deviation, minimum and maximum, keeps the last row when a date is duplicated in a summary file, merges dates across files and writes to staging, converting unavailable numbers to null. A separate route attaches a survey file with date, parsed send time and stress columns, sorts by time and takes the last response of each date. That route updates only dates already in staging. The reveal route promotes one date at a time, as described in Chapter 4, and re-uploading sets the revealed flag back to false for matching rows, so repeated upload is not harmless on a dataset already replayed.

5.3 Integration with the Other Components

PULSE shares four things with the rest of CareSense. The authentication service and its profile roles are common. The sidebar and navigation shell list all four modules, and PULSE appears there as Deterioration Detection beside Fall Detection, Sign and Vitals, and Voice Log. An audit log table is defined once and shared. A model registry records the component, name, version and metrics for every module that has a model. Everything else, including PULSE's tables, routes and models, belongs to the module.

Final integration of the four components is complete, and the integrated application runs.

5.4 Verification and Software Testing

5.4.1 Strategy

Verification worked at three levels. The first checked the research pipeline for leakage and instability. The second checked the model-serving path, meaning that stored

features produce a score through the service. The third checked application behavior against the requirements of Chapter 4.

5.4.2 Research pipeline checks

- The saved allocation was loaded and not recreated, and the fifteen demonstration people were confirmed absent from the research table.
- No person appears in more than one group, since allocation was made at the participant level.
- Imputation, class weight and both models were fitted on training people only, and early stopping used validation AUC.
- A regularized run replaced an unregularized one after a large gap between training and test AUC.
- A strictly historical rebuilding of the steps feature was run as a sensitivity analysis.
- Repeated participant-grouped cross-validation checked that the result was not an accident of one split.

These checks reduce specific risks. The test group was inspected while the model was being developed, the previous-response chronology rests on row order and the original surveys' completion times were not checked.

5.4.3 Model-serving and application checks

A batch script authenticates as an administrator, checks the health route, resolves participant identifiers to caregiver profiles, enumerates the historical dates and calls the scoring route for each. It wrote 1,487 rows, each with probability, a class and a top factor. The file has no true labels, so it shows that the serving path runs end to end on historical rows and does not measure accuracy. Beyond it, I retain screenshots of the main views and confirm that the current application versions run without known unresolved faults in PULSE.

Table 5.2 gives the test cases. It is a specification of the scenarios the requirements call for, together with an honest record of the evidence kept for each. Some cases rest

on a retained screenshot or output. Others rest only on my confirmation of working behavior.

Table 5.2: Test cases, requirements covered and evidence retained

ID	Covers	Scenario	Expected result
TC1	FR1, NFR1	Administrator signs in and calls /me	Role returned as administrator
TC2	FR1, FR2, NFR1	Supervisor requests a caregiver outside the team	Expected answer returned
TC3	FR2	Administrator lists monitored caregivers	All monitored caregivers listed
TC4	FR4, NFR3	Score a chosen date	Probability, class and ordered factors returned
TC5	FR4, NFR4	Score a date with missing features	Nulls returned, no failure
TC6	FR5	View risk history containing unscored days	Gaps shown, not treated as low risk
TC7	FR6	View expected against actual heart rate	Both series plotted
TC8	FR7	View summary, heatmap and trends	Counts and charts match stored scores
TC9	FR9, FR10	Flag a caregiver with sustained risk and an eligible same-unit candidate	Pending recommendation with the lowest-risk candidate
TC10	FR9, FR13	Flag a caregiver with no eligible candidate	Unmatched recommendation with reasoning
TC11	FR9	Caregiver with fewer than five scored rows	Not flagged
TC12	FR9	Caregiver already holding a pending recommendation	No duplicate created
TC13	FR10	Mark a recommendation reviewed	Reviewer and time recorded; roster unchanged
TC14	FR11, NFR1	Non-administrator calls an upload route	Checked and validated
TC15	FR11, NFR4	Upload a file lacking a required column	400 returned with a message
TC16	FR11	Administrator uploads valid files	Dates staged
TC17	FR12	Reveal days one at a time	No score for days 1 to 7; score from day 8
TC18	NFR5, NFR9	Start the service with a model file missing	Start-up fails with an error
TC19	NFR2	Inspect the candidate fields in a recommendation response	Only the fields needed for review
TC20	NFR8	Open PULSE from the shared sidebar	Module loads inside the shell

5.5 Deployment and Operation

The application ran locally when it was shown at the progress presentation. The service was started with Uvicorn on port 8000, and the front end development server on port 5173, with cross-origin access permitted for that one origin only. The three model artifacts were placed in the models folder before start-up.

The application was deployed on Render and Vercel before the final presentation, with the FastAPI service on Render and the React front end on Vercel. The database and authentication remained on Supabase, and the service-role key and other secrets were set as environment variables on the host and were not placed in the source code. The cross-origin setting was changed to allow the front-end address of the hosted site. The earlier model files were replaced with the final research models, and I verified the replacement by comparing file hashes, feature order, preprocessing and the predictions on a frozen set of rows, within a stated numerical tolerance.

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 Development sequence and the overfitting correction

Table 6.1 traces the path from the first same-day models to the final pipeline, and Figure 6.1 plots it. The early runs sat close to chance. Rebuilding features at the survey cut-off gave 0.5361 with fifteen inputs, and adding the previous stress response raised the test AUC to 0.6526. Regularization brought it to 0.7086, and replacing the flat personal baseline with the learned expected heart rate gave 0.7211.

Table 6.1: Development sequence of test ROC AUC

Step	Input or change	Test ROC AUC
Same-day full sample	Retrospective individual features; separate diagnostic run	0.5081
Pre-survey features	Fifteen features including sleep; no previous stress	0.5361
Add previous stress	Sixteen features; unregularized classifier	0.6526
Regularize and stop early	Shallower trees, penalties and a flat historical baseline	0.7086
Learned expected heart rate	Stage 1 regressor replaces the flat deviation; final classifier	0.7211

Source: Author's notebook outputs.

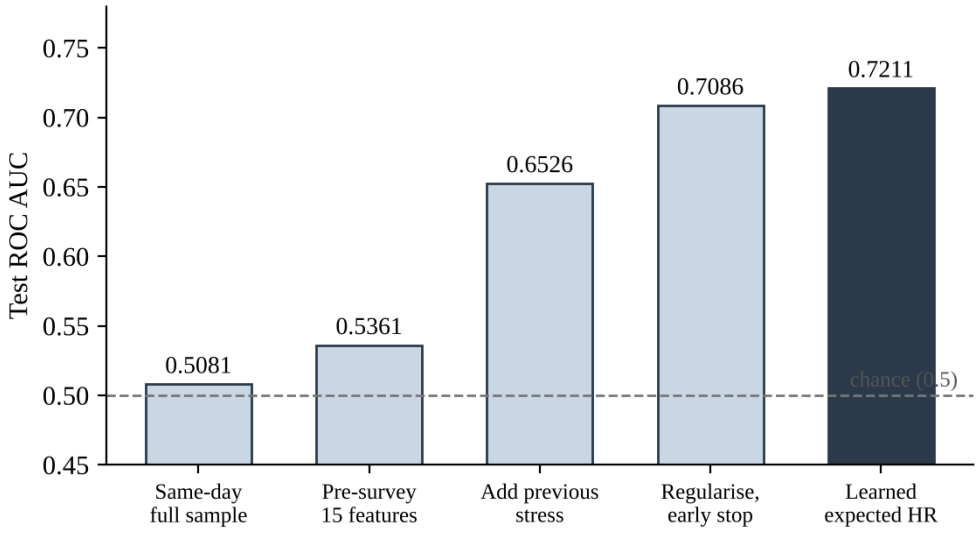


Figure 6.1: Test ROC AUC across the development sequence

Preprocessing and row eligibility changed between them, so the differences cannot be attributed to a single change. The record also shows repeated use of the test group during development, which is why the thesis calls the final estimate a good one for these 27 people and not an untouched confirmation.

The overfitting diagnosis deserves its own table. Table 6.2 compares training, validation and test AUC across the three classifier runs that used the previous response. The first run scored 0.8654 on training rows and 0.6526 on test rows, a gap of 0.2128. Regularization reduced the gap to 0.0172. The final run, with the learned baseline, scored 0.7328, 0.6874 and 0.7211.

Table 6.2: Training, validation and test ROC AUC across three classifier runs

Run	Training	Validation	Test	Train–test gap
Unregularized lag model	0.8654	0.6855	0.6526	0.2128
Regularized, flat baseline	0.7259	0.6970	0.7086	0.0172
Final, learned baseline	0.7328	0.6874	0.7211	0.0117

6.1.2 Final held-out performance

Table 6.3 reports the final results on the 27 held-out caregivers and their 1,406 usable surveys. The test ROC AUC was 0.7211, with a 95% caregiver-bootstrap interval of 0.6644 to 0.7699. Accuracy at the default threshold of 0.5 was 0.7930, with an interval of 0.7265 to 0.8591.

Table 6.3: Final results on the held-out test group

Result	Value
Test population	27 caregivers; 1,406 usable surveys
Training ROC AUC and accuracy	0.7328 and 0.7691
Validation ROC AUC and accuracy	0.6874 and 0.6966
Test ROC AUC (95% CI)	0.7211 (0.6644 to 0.7699)
Test accuracy (95% CI)	0.7930 (0.7265 to 0.8591)
Share of test rows labelled elevated	0.2127
Accuracy of always predicting not elevated	0.7873
Final boosting rounds	11

The accuracy figure needs care. A model that said “not elevated” every time would have scored 0.7873 on these rows, so the full model beats that baseline by only 0.0057,

or about half a percentage point. The AUC is the better evidence of ranking ability, and it indicates moderate discrimination. It does not support a claim of clinical reliability, burnout prediction or care-home effectiveness.

6.1.3 Uncertainty and repeated validation

The repeated participant-grouped validation produced twenty-five fits on development people, with a mean AUC of 0.6991 and a standard deviation of 0.0294. The mean sits a little below the test estimate, which is what one would expect if some of the test result reflects favorable luck of the draw, though the evidence cannot prove it. The spread shows that results vary noticeably from one set of held-out people to another. This supports stability within the TILES development population and says nothing beyond TILES.

6.1.4 Incremental value over the previous stress response

On the 1,379 of 1,406 test rows that had a previous response, the full model scored an AUC of 0.7273 and the previous response alone scored 0.7026. The difference of +0.0247 had a paired caregiver-bootstrap interval from −0.0010 to +0.0515. The interval includes zero. Most of the signal therefore comes from the previous response, and a clear additional gain from physiology, activity and sleep has not been established. The two AUC values are computed on a smaller population than the full-test figure, so they should not be compared with 0.7211 directly.

6.1.5 Sensitivity to a strictly historical steps baseline

Rebuilding the steps deviation from earlier dates only, and refitting Stage 1, the imputer and Stage 2 with the same settings, gave a test AUC of 0.7213 with an interval of 0.6650 to 0.7692. The result is almost the same as the main run. That shows the exception did not drive the finding in this experiment. It does not turn the main run into a prior-only pipeline.

6.1.6 Proximity centrality

The matched experiment compared a model without network features, Model A, with one that added degree and weighted centrality, Model B, on 2,989 rows that had network values. Table 6.4 and Figure 6.2 give the change in test AUC for each model

family. Every difference was small and every interval included zero. Adding centrality produced no detectable improvement in any family.

Table 6.4: Matched comparison of models with and without centrality

Model family	A without network	B with network	Difference (95% CI)
Random forest	0.4576	0.4548	−0.0028 (−0.0379 to +0.0361)
XGBoost	0.4604	0.4460	−0.0144 (−0.0506 to +0.0269)
LightGBM	0.4434	0.4345	−0.0089 (−0.0487 to +0.0307)

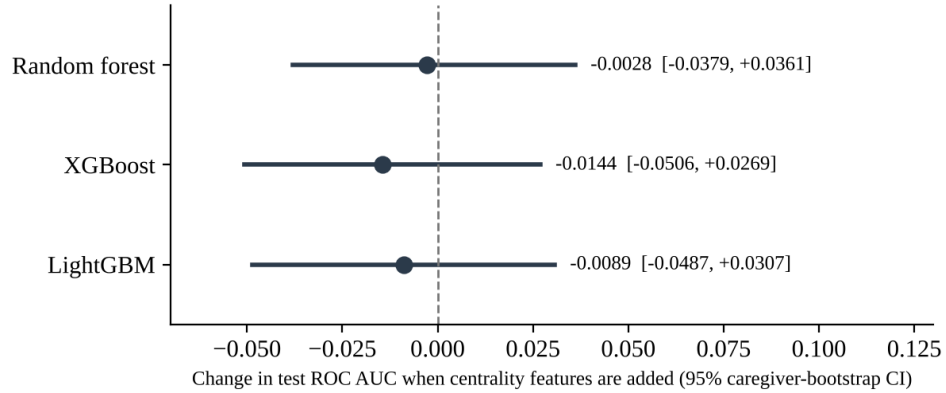


Figure 6.2: Change in AUC when centrality is added, by model family

Two cautions apply. The base AUC values are below 0.5 in these runs, which means the matched models ranked the labelled rows no better than chance, and the comparison therefore says little about what a good individual model could gain from network features. Coverage was limited to 29.8% of the retrospective rows, and the location signal is noisy. The finding is that no benefit was detected in this experiment. It is not a finding that colleague contact is irrelevant or harmful.

6.1.7 Explanations

The explanation of the final classifier used the exact tree explainer on the imputed test matrix. Figure 6.3 shows the mean absolute contribution of the ten largest features. The previous stress response dominates, at 0.2080, followed by the highest heart rate of the day (0.0678), the seven-day historical deviation (0.0465) and the heart-rate spread (0.0215). Sleep efficiency and fat-burn minutes come next at about 0.01 each. The learned heart-rate residual contributes only 0.0031 on average.

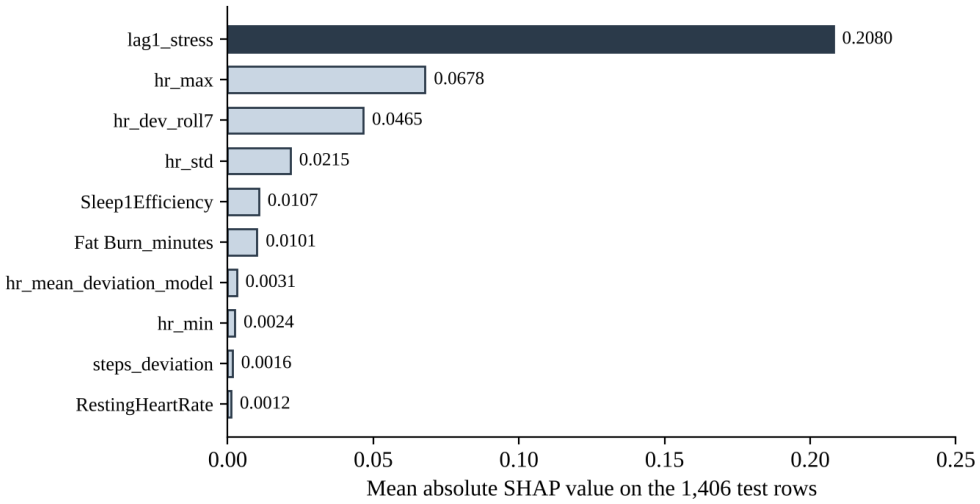


Figure 6.3: Mean absolute SHAP value of the ten largest features on the test rows

Six features received no attributed contribution in this fitted model: the mean heart rate, the three-day deviation, step count, out-of-range minutes, cardio minutes and peak minutes. A zero describes this fitted run and does not prove that the feature has no physiological relevance. The original summary plot produced by the notebook is reproduced in Figure 6.4.

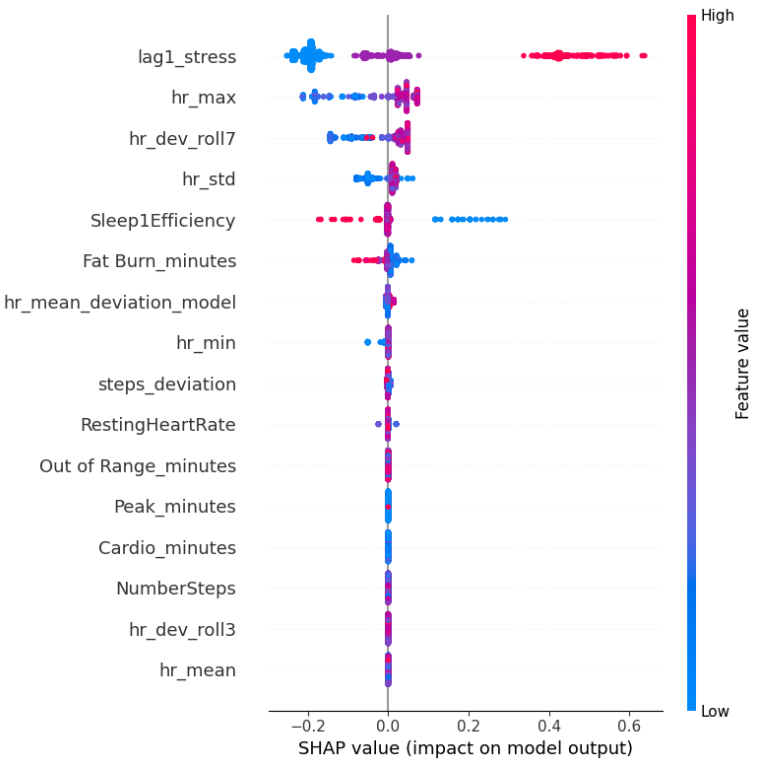


Figure 6.4: SHAP summary plot from the final notebook output

6.1.8 Exploratory associations

On the retrospective table, the personalized heart-rate deviation correlated weakly with the label ($r = 0.024$, $p = 0.017$), the raw heart rate did not ($r = 0.005$, $p = 0.625$), and sleep efficiency correlated negatively ($r = -0.106$, $p < 0.0001$). A median split on centrality gave a Welch t of −1.5172 ($p = 0.1294$, Cohen's $d = -0.061$), and a workload interaction gave $p = 0.066$.

6.1.9 Supplementary results without the previous stress response

Table 6.5 gives the matched version 1 (V1) comparison on the 1,406 test rows. Without the previous response the AUC fell to 0.6108, against 0.7213 with it. The paired bootstrap difference was −0.1105, with an interval of −0.1774 to −0.0454, so the loss is clear. The apparent strength of the no-response model at the default threshold is misleading. It labelled every one of the 1,406 test rows as not elevated, so its 0.7873 accuracy reflects the class balance and nothing more. The sixteen-feature comparator detected only about 5.35% of the elevated cases at the same threshold, despite its better ranking.

Table 6.5: V1 matched comparison on the test group

Metric	No previous response (15 features)	With previous response (16 features)
ROC AUC	0.6108	0.7213
PR AUC	0.2919	0.4068
Brier score	0.1739	0.1691
Accuracy at 0.5	0.7873	0.7930
Precision at 0.5	0.0000	0.6667
Recall at 0.5	0.0000	0.0535
F1 at 0.5	0.0000	0.0991

Table 6.6 summarizes all five supplementary experiments and, importantly, the cohort each one used. Version 2 (V2), Version 3 (V3) and Version 5 (V5) report validation results, so their AUC values must not be set against the V1 test figures as if both measured the same people. The best validation AUC across V2 and V3 was 0.5623 and 0.5661 respectively, a difference of 0.0038 that does not establish an improvement. Version 4 (V4) produced descriptive differences only. Within caregivers, the median short-window heart rate was higher in surveys rated elevated, by about four beats per minute for the fifteen-minute window, although about a quarter or more of eligible caregivers did not show that direction. V5 then tested whether seventeen new features built on those windows improved prediction, and they did not. The validation AUC was 0.5572 for the baseline and 0.5360 for the enhanced set. Threshold-level detail for V2, V3 and V5 is in Appendix A.

Table 6.6: Summary of the supplementary experiments

Version	Question	Cohort evaluated	Principal result
V1	What happens without the previous response?	Test, 27 caregivers	AUC 0.6108 against 0.7213; clear loss
V2	Can model choice and tuning recover performance?	Validation, 30 caregivers	Best AUC 0.5623 across 24 configurations
V3	Do other model families or engineered features help?	Validation, 30 caregivers	Best AUC 0.5661; the 26-feature set did not win
V4	Do raw short-window heart-rate samples differ within people?	Training only, 132 caregivers	Descriptive within-person differences; no classifier trained
V5	Do seventeen new short-window features improve prediction?	Validation, 30 caregivers	AUC 0.5360 against 0.5572; no improvement

The threshold behavior in V2 and V3 is a warning. When a threshold was chosen to maximize F1 on validation data, the classifier flagged about 96.5% of assessments in V2 and about 95% in V3, catching nearly every elevated case by flagging nearly everyone. At a threshold of 0.5, the same V3 model identified one elevated assessment in 393. A score that can behave in either extreme depending on a threshold is not ready to drive any alert. The overall conclusion of the supplementary work is that no questionnaire-independent replacement was found with the methods tried.

6.2 Product and Technical Results

The screenshots in this section come from the current application and use anonymized caregiver identities backed by historical readings. They show what the interface displays.

Figure 6.5 shows the Risk Analysis view for one assessment. It presents the estimate, the factors that raise and lower it, and a history chart with a heart-rate overlay. Figure 6.6 shows the personal baseline view, which compares expected and actual heart rate. Figure 6.7 shows the directional factor lists. In all three, the estimate is labelled as a model estimate and the color encodes a display tier.

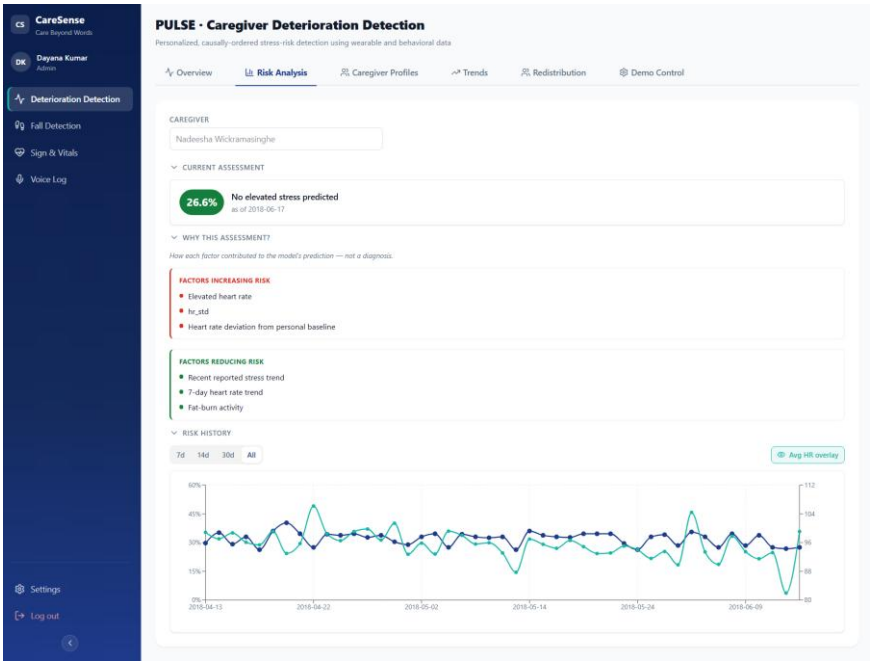


Figure 6.5: Risk Analysis view with an estimate, directional factors and history

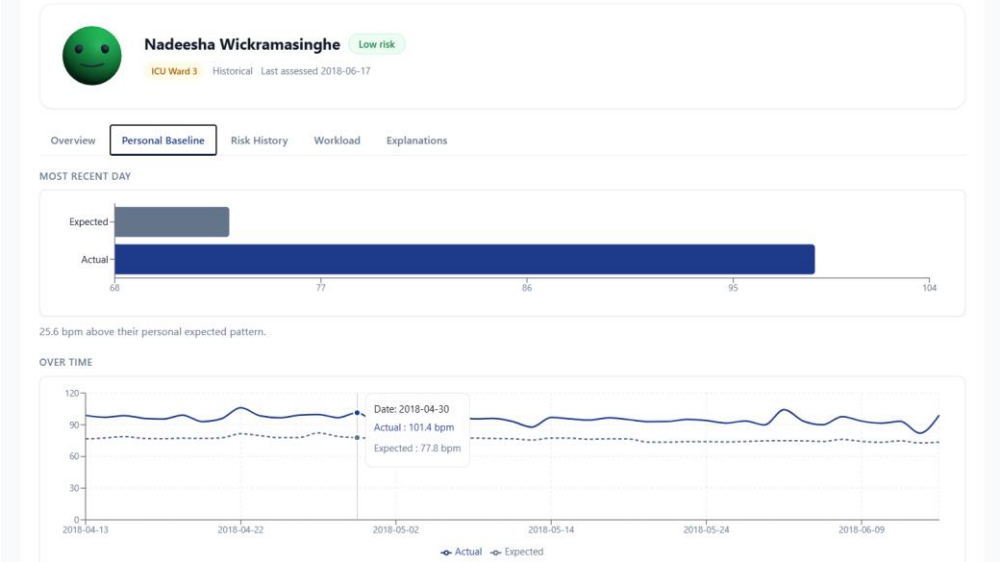


Figure 6.6: Personal baseline view comparing expected and actual heart rate

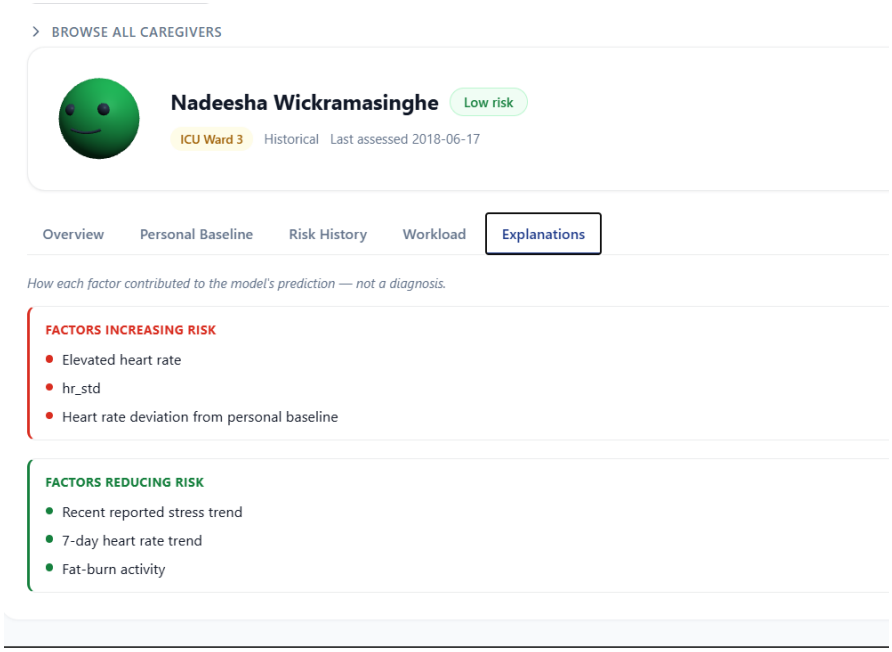


Figure 6.7: Directional factor lists for one assessment

Figure 6.8 shows the redistribution tab. In this demonstration the rule evaluated the historical caregivers and flagged two. One was matched with a lower-risk colleague from the same unit, and the other was reported as unmatched, with the reason that no colleague in the unit currently had low enough risk. A review button records acknowledgement. Figure 6.9 shows the administrator's upload and replay controls.

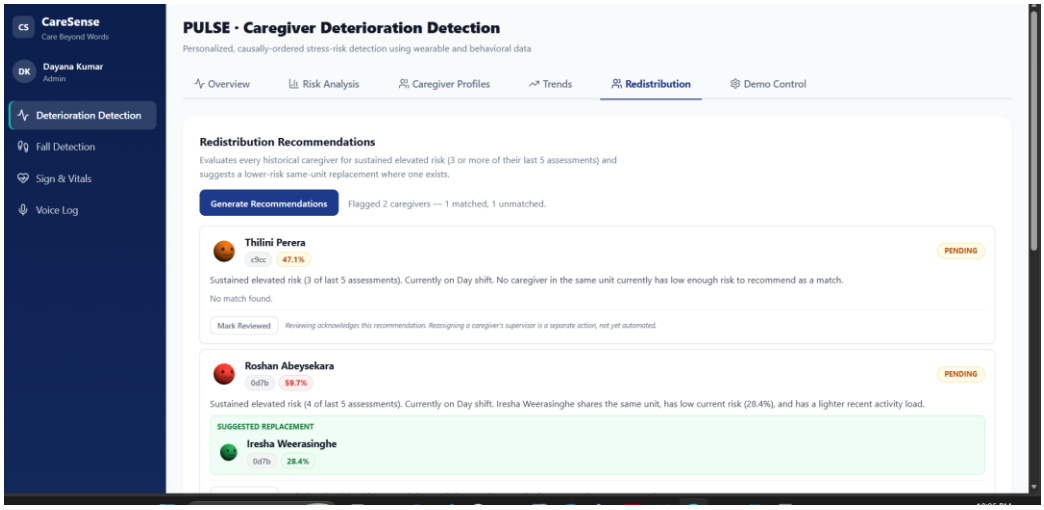


Figure 6.8: Redistribution recommendations with one matched and one unmatched case

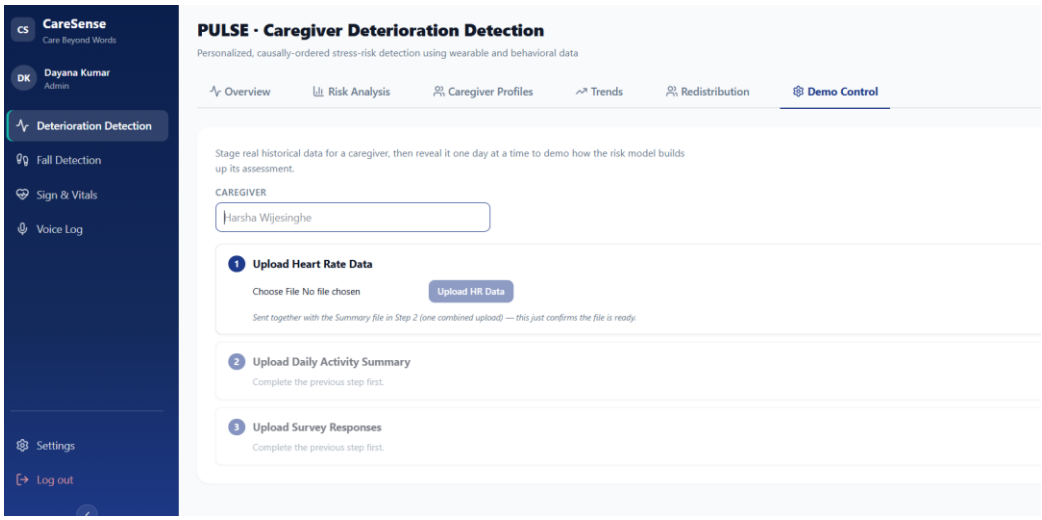


Figure 6.9: Administrator upload interface for staged historical readings

The batch scoring run produced 1,487 historical scores through the service, which shows that the serving path works across many dates. The output has no true labels and so provides no accuracy figure. The application displays model output as a percentage, calculated by multiplying the probability by 100.

6.3 User or Stakeholder Validation

The application was demonstrated to two stakeholders from the target setting: a representative of Remark Elder Care and a caregiver working in elder care. I presented the working prototype using the demonstration group of 15 participants, whose data were kept out of model development, and both reviewers were shown the overview, the risk analysis, the caregiver profiles, the trends and the redistribution suggestions. The sessions were walkthroughs followed by open discussion, with no questionnaire or timed task, so the findings are qualitative. The Remark Elder Care representative advised that a suggested substitute should reflect the skill mix, shift pattern and minimum staffing level of a unit, and not only a low score. The representative also advised that the score should start a conversation with a caregiver and not serve as a judgement, that caregivers should be informed and should consent before real monitoring begins, and that a trial with data from the care home would be needed before the scores could be relied on.

The caregiver found the tier colors and the list of factors behind each score easy to follow, and noted that a percentage could be mistaken for a medical result. This supports the choice to describe the figure as a model score. The caregiver also asked that a flagged person be able to see why they were flagged and to explain a high score, since a busy shift or an emergency can raise heart rate, and noted that the number and condition of residents on a shift would add useful context. I recorded each comment as a requirement or a limitation.

6.4 Integration Results

PULSE is integrated into CareSense through four shared parts: the authentication service with its profile roles, the sidebar and navigation shell, the audit log, and the model registry. Final integration of the four components is complete, and the integrated application runs. Figure 6.10 shows the integrated sidebar with all four modules, in which PULSE appears as Deterioration Detection beside Fall Detection, Sign and Vitals, and Voice Log. A user signs in once, and the same token and profile role then decide what that user may open in every module. In PULSE this role feeds the scope checks described in Chapter 4, so a supervisor sees only their own team in the integrated application, as in the stand-alone one. The module-specific parts did not need to change when the other components were added. The PULSE routes, tables and models stayed as they were, which is the intended result of keeping the boundaries narrow. The other modules' data and models also remain separate from PULSE.

The shared model registry records the component, model name, version and metrics for every module that has a model, so the integrated application can show which model produced a score. The integration was confirmed by running the application with all four modules.

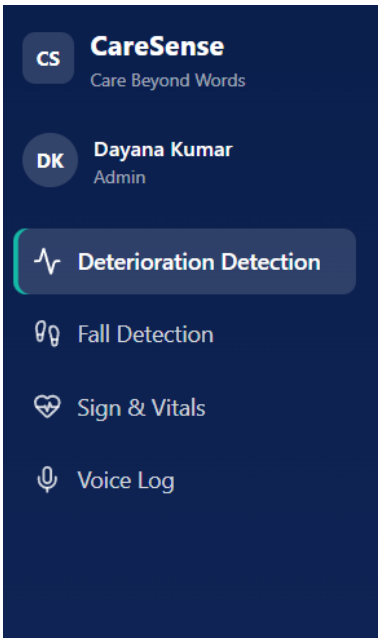


Figure 6.10: Integrated CareSense sidebar showing the four modules, with PULSE as Deterioration Detection

6.5 Summary of Findings

Table 6.7: Summary of findings by research question

Question	Finding	Strength of evidence
RQ1	Moderate ranking of elevated stress in held-out caregivers: AUC 0.7211 (0.6644 to 0.7699); repeated validation mean 0.6991	Within-cohort; test group inspected during development
RQ2	No detectable gain from proximity centrality in any of three model families	Limited network coverage; base AUC below 0.5 in the matched runs
RQ3	The full model beat the previous response alone by +0.0247, interval −0.0010 to +0.0515	Interval includes zero; no clear gain established
RQ4	Strictly historical steps baseline gave AUC 0.7213, almost identical to the main run	Single sensitivity run
RQ5	Explanations, sustained-risk histories and a reviewable match are implemented and demonstrated	Demonstration only; usefulness not evaluated

7 DISCUSSION

7.1 Interpretation of Findings

The aim of this research was to estimate the probability of elevated stress in a caregiver from wearable data and the earlier survey response, to test the estimate on people the model had not seen, and to present it in a way that a supervisor can review. The findings meet that aim, and I discuss each research question in turn.

For RQ1, the final pipeline ranked elevated stress in unseen caregivers better than chance. The test AUC was 0.7211 (95% CI 0.6644 to 0.7699), and the repeated grouped validation gave a mean of 0.6991 with a spread of 0.0294. The two figures agree once that spread is allowed for. Together they show moderate ranking ability, which is what a self-reported rating given at an unpredictable moment, with a few wearable signals, might be expected to support. The learned expected heart rate also helped slightly: replacing a flat personal mean raised the AUC from 0.7086 to 0.7211. This supports the personalized design. Accuracy needs care. The model reached 0.7930, but a rule that always predicts “not elevated” reaches 0.7873. Elevated stress is the minority outcome, so accuracy mostly measures how often the majority class occurs, and I judged the models by AUC.

For RQ3, the previous stress response carried most of the signal. SHAP gave it the largest contribution (0.2080). The full model beat the previous response alone by 0.0247 in AUC, but the interval ran from −0.0010 to +0.0515 and includes zero. The wearable features may therefore add a little information. The experiments run without the previous response point the same way. Removing it lowered the test AUC from 0.7213 to 0.6108, and later attempts with other algorithms and added heart-rate features reached validation AUC values of only about 0.56. In this dataset, with these methods, wearable and activity signals alone ranked stress only slightly better than chance. The module is therefore best suited to settings where the same people keep answering short check-in surveys, and less suited to passive monitoring alone.

For RQ2, proximity centrality has nothing detectable. The change in AUC was −0.0028 for the random forest, −0.0144 for XGBoost and −0.0089 for LightGBM, and every interval crossed zero. Two limits apply. The matched models ranked rows no

better than chance to begin with, so there was little for the network to improve, and only 29.8% of the retrospective rows had network values. The result means that this construction of the network, on this subsample, gave no benefit. It does not mean that social contact is unimportant for staff stress.

For RQ4, rebuilding the steps-deviation feature from strictly earlier information gave an AUC of 0.7213, almost identical to the main run. The one feature that was not strictly historical therefore does not appear to explain the result. This is one check on one feature, so it does not rule out other timing effects.

For RQ5, the application shows that the estimate can be presented in a form a supervisor can review. It gives tiered scores, the factors behind each score, a trend with a heart-rate overlay, and a redistribution suggestion that needs several elevated results in a row and a human decision.

Taken together, the findings agree with the aim in one respect. The pipeline can rank elevated stress in people it has not seen, the previous survey response explains most of that ability, and the added value of wearable and network features remains unproven.

7.2 Comparison with Literature Baselines and Existing Solutions

I did not compare the AUC of 0.7211 with scores reported in other studies, because the studies differ too much for the numbers to mean the same thing. They use different people, different labels, different time windows and different sensors. Most of all, they differ in how the test people were kept apart from the training people. Cahoon and Garcia reached a similar view when they examined how well stress monitoring for healthcare workers carries over between real-world datasets [12]. A comparison of the approaches is therefore more useful, and I make it below.

Early work on stress detection from body signals reported high accuracy, but under controlled conditions such as driving tasks and laboratory protocols [16], [24]. In those studies the stress label comes from a known task. In this study it is a single rating that a hospital worker gave at an unpredictable moment of the day, and the test people were never seen in training. This is a harder problem, so a lower score is expected. Saeb and

colleagues argue that a study should be tested in the way the model will be used, and that mixing a person's data across training and test sets gives results that look better than they are [8]. The person-level split, the repeated grouped validation and the bootstrap over caregivers in this study follow that advice.

Some studies have worked with healthcare staff and used passive data. The HiPAL study built a deep learning framework to predict physician burnout [27], and Rana and colleagues reported passive detection of stress and burnout among frontline workers [26]. Both address outcomes related to the one here, with different data and labels. Three points set PULSE apart. It records the time of every input, so no reading comes from after the survey it predicts. It reports a plain comparator, the previous survey response, next to the full model, which shows how much of the result that one answer explains. It also tests proximity against a matched model without it, and does not assume that proximity helps.

Existing workforce tools fall into two groups, as Section 2.3 described. Rostering systems plan cover and satisfy constraints, but they do not consider how strained an individual is [23]. Wellbeing and wearable dashboards show personal measures, but they do not link them to a staffing action. PULSE sits between the two. It estimates the probability of elevated stress and suggests a colleague in the same unit, and a person makes the decision. It is a research prototype and should be judged as one.

7.3 Research Contribution

This research makes four contributions to the study of stress prediction in healthcare workers, and each addresses a gap identified in Section 1.4.

The first is a reproducible method for building and testing a stress model on workplace data. Participants are allocated to training, validation, test and demonstration groups before any feature is built, and the allocation is saved and reused in every later run. Every input is tied to a time earlier than the survey it helps to predict. All fitted steps, including imputation and class weights, use the training people only. These rules are written into one consolidated notebook that rebuilds the analysis from the raw files. Other researchers working with TILES-2018, or with similar datasets, can follow the same steps and check their results against the figures reported here.

The second is a two-stage design that personalizes the prediction. The first stage learns the heart rate expected for a caregiver at a given time, and the second stage works with the difference between the expected and the observed value. This replaced a fixed personal average and raised the AUC from 0.7086 to 0.7211. The gain is small, but it is consistent, and it shows that the context of a person's day can be modelled without training a separate model for each person. For people with few labelled surveys, as in this dataset, that is a practical advantage.

The third is a set of paired and matched comparisons that show how much each part of the model adds. The previous survey response was compared with the full model, proximity features were compared with a model without them on identical rows, and a strictly historical version of one feature was compared with the main version. Each comparison reports an interval calculated over people. This allows a reader to see which components carry the result. The main finding is that the previous response holds most of the signal, and that the gain from wearable features is small and uncertain. The same finding applies to proximity centrality in three model families. Studies of this kind seldom report such comparators, and they should be included in future work so that the contribution of passive sensing can be judged.

A further contribution is the set of experiments run without the previous response. They were carried out with several algorithms, added heart-rate features and an examination of the raw readings, and none gave a useful replacement. These results save other researchers from repeating the attempt on the same data, and they define where the approach works: when caregivers answer short check-ins regularly. Together with the application described in Chapter 4, which shows the estimate to a supervisor with its reasons and keeps a person in charge of any staffing decision, the research offers a tested starting point for later work in care settings.

7.4 Product Contribution

The product contribution of this work is PULSE, a working decision-support module that turns the research model into something a supervisor can use. It gives the supervisor an estimate of the probability of elevated stress for each caregiver, the reasons behind it, and a way to act on it that keeps a person in charge.

The module is organized around the questions a supervisor is likely to ask. The overview shows how many caregivers fall in each tier and which of them need attention, and a heatmap shows the team as a whole. For a single caregiver, the risk view shows the probability, the factors that raised it and the factors that lowered it, and a history chart places the score beside the expected and the observed heart rate. The profile and trend views show how the pattern changes over time. A supervisor does not need to read model output to follow any of this. The tiers, the plain-word factor names and the wording of "model score" were chosen so that the figure is not mistaken for a medical result.

The second part of the contribution is the way a score becomes an action. A single high result does not raise a flag. A caregiver needs at least five scored assessments, and three or more of the last five must be predicted elevated. The module then searches for a colleague in the same unit with a lower recent score, and it states plainly when no suitable colleague exists. The suggestion is stored with its reasoning and is reviewed by a person, and the module itself changes no roster. The first seven days of a replay are marked as unscored, so that a missing score is never read as a low one. These choices keep the module within what the evidence supports, and they show a way of connecting a prediction to a staffing decision without handing the decision to the model.

The module is also built to be used inside a larger system. It shares authentication, the sidebar, the audit log and the model registry with the other three CareSense modules, while its routes, tables and models stay its own. Supabase supplies the sign-in, the roles and the row-level security, and the service checks the role and scope of every request, so a supervisor sees only their own team. Because the data are historical, the application includes a replay that presents them one day at a time, and this allowed the

module to be demonstrated to a Remark Elder Care representative and a caregiver without live sensor data. Both could follow the screens, and their comments shaped the list of future work.

7.5 Limitations and Threats to Validity

The limitations are set out here so that the reader can judge the results against them. They are grouped by the type of validity each one affects: construct, internal, external, and product and evaluation.

7.5.1 Construct validity

The outcome in this study is a self-reported stress rating. A caregiver answered a short question on a five-point scale, and I counted a rating of three or more as elevated stress. The rating describes how the person felt at the moment of the survey. It does not measure burnout, depression or any other clinical state, and the study makes no claim about them.

A single rating is also a noisy measure. People interpret the scale in their own way, and one answer given in the middle of a shift may not reflect the person's usual state. Some of the variation in the outcome is therefore measurement noise that no model could explain. The wearable inputs add a similar limit. Heart rate and activity are indirect signals of stress, because they also respond to exercise, caffeine and posture.

7.5.2 Internal validity

The test group of 27 people was examined repeatedly during development. The final estimate is therefore a fair measure for these 27 people, but it is not an untouched confirmation on new ones. The repeated grouped validation partly offsets this, since it tests the pipeline on many different splits. It also used the training and validation people, so it does not replace a fresh test group.

The development steps also changed several things at once, such as the preparation of the data, the rows kept and the features used. The gains between one step and the next

cannot be assigned to a single cause. Two features need a note. The lag feature relies on the previous usable response, which may be hours or days earlier.

7.5.3 External validity

The data come from hospital workers in the United States. Care-home staff in Sri Lanka or the United Kingdom work under different conditions, with different shifts, tasks and resident needs. The participants were also volunteers who agreed to wear sensors and answer surveys for about ten weeks. People who chose to take part, or who stayed in the study, may differ from the wider workforce.

7.5.4 Product and evaluation limits

Any real deployment of this module would carry privacy and data-protection duties. In Sri Lanka, the Personal Data Protection Act would apply [30]. Staff data could be used only after ethical and organizational approval, and caregivers would need to be told what is monitored and to agree to it. The current work uses a licensed research dataset and synthetic display identities, so none of these duties arose in this study, but they would have to be met before the module could be tried in a care home.

7.6 Implications

The findings have implications for how a module like PULSE should be used in a care setting, for how similar studies should be reported, and for the next stage of this project.

For practice, the findings point to one kind of use. The module works best when caregivers answer short check-in surveys regularly and the answers are combined with wearable data, because the previous response carried most of the signal and the wearable signals alone ranked stress only slightly better than chance. The scores should be treated as prompts for a conversation with the caregiver. A supervisor who read a high score as a diagnosis, or used it to judge performance, would be misusing the tool. The reasons shown beside each score and the human review of every suggestion exist to prevent that reading. Research on interpretable models supports this design, because it argues that an explanation should be judged by what it lets a person do with the result, and not by how technical it looks [20].

For research, two points follow. Studies of workplace stress prediction should report a simple comparator such as the previous response, because without it the contribution of passive sensing cannot be judged. A reader who sees only the full model's score cannot tell whether the sensors added anything. A proximity network should also be compared against a matched model without it before it is described as useful, since in this study it gave no detectable gain. Both are cheap to add to a study and make its claims easier to trust.

For the next stage of this project, the most useful step is a small, ethically approved evaluation with real reviewers, and, if permitted, data from the target care settings. The first would show whether supervisors find the presentation useful and clear. The second would show whether a model built on hospital workers holds up in a care home.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

PULSE is meant for the people who plan and supervise care work in residential care settings. As Section 1.3 explains, a supervisor usually hears about a caregiver's strain through conversation or observation, and often only after the work has been affected. PULSE offers an earlier and more regular view. For each caregiver it gives an estimate of the probability of elevated stress, the factors that moved the estimate, and a suggested colleague from the same unit who could take over some work. A person reviews every suggestion.

Three groups are involved. Supervisors and administrators are the direct users, since they work with the views described in Chapter 4. Caregivers are the people whose data are used. The benefit to them is a fairer spread of work, but they are also the group that could be harmed if the estimates were misused, which is why consent and transparency matter so much (Section 3.6). The third group is the care home operators, who would pay for the system. Staff shortages, burnout and unsafe staffing are linked with poorer care [6], so an operator has a practical reason to want an early warning.

The value claimed here is a design value. PULSE gives a supervisor a transparent estimate that can be questioned and overruled.

8.2 Market and Competitive Position

I carried out a desk-based market review, using published market reports, national statistics and the studies reviewed in Chapter 2. The figures below describe the market around PULSE.

PULSE sits between two commercial markets. The first is healthcare workforce management, which covers scheduling, talent management and analytics. One industry report values it at 1.7 billion US dollars in 2024 and expects 2.8 billion by 2029, a yearly growth of 10.1% [32]. The report names nursing homes as one group of end users, but it gives no separate figure for them. The second market is wearable health devices, valued at 41.07 billion US dollars in 2024 and expected to reach 75.98 billion

by 2030 [33]. These are global estimates from commercial research firms and I could not check them independently. I use them to show the direction of the market and not its exact size.

Demand comes from the shortage of care staff. The World Health Organization projects a shortfall of 11 million health workers by 2030 [34]. Sri Lanka is ageing quickly. The 2024 census found that 12.6% of the population was aged 65 or over, compared with 7.9% in 2012 [35]. In June 2026 the Minister responsible reported 455 elders' homes across the country, of which only 148 were registered [36]. That gives a rough idea of the first group of possible customers: a few hundred homes, of which about a third are registered.

The alternatives were reviewed in Section 2.4. Supervisors rely on conversation, observation and occasional surveys. Consumer wearables report stress or wellness scores to the wearer and do not turn the score into a staffing step. Rostering systems build schedules from rules and availability [23], and they do not consider how strained each person is. Research prototypes have addressed burnout among physicians using records from electronic health systems [27] and among nurses through a mobile intervention [28]. These are aimed at hospital clinicians as individuals, and none of them is a tool that a care home supervisor can buy and use.

The position PULSE aims for is a small or medium care home that wants a transparent tool and not a hidden score. Its strengths are the explanation shown with every estimate, the human review of every suggestion, and its place inside CareSense, where a user signs in once and the actions are recorded in a shared audit log. Its weaknesses are plain. The held-out AUC of 0.7211 is moderate, most of the signal comes from the previous stress response, and the model was trained on hospital staff. The sensible starting point is a supervised pilot in a few care homes, and the module should be retrained and tested on care home staff before it is offered as a finished product.

8.3 Business or Sustainability Model

The model below is a proposal. The most natural arrangement is for PULSE to be licensed together with the other CareSense modules, as a yearly subscription for each care home site. The subscription would cover hosting, support and updates to the model. A first step would be a pilot in a single home, with the consent of staff, so that the module can be tried and local data can be gathered for retraining.

The costs I can identify are cloud hosting for the backend and the database, the wearable devices, the time of a support person who sets up the system and trains supervisors, and the ethics and legal review that Section 8.4 describes.

The model also has to stay useful over time. Staff, routines and seasons change, and a model that was trained once will slowly fit the care home less well. A sustainable plan would retrain it on local data using the same checks as in this thesis, with people kept separate between training and testing, and would record each version in the model registry that CareSense already provides (Section 6.4).

8.4 Scalability, Intellectual Property and Compliance

On scalability, the backend is a FastAPI service that reads from a Supabase PostgreSQL database, and access is limited by role and by unit. In principle, adding a care home means adding records and units and does not require new code. A larger gap is that the application loads historical records through an administrator's upload and replay. A real deployment would need an automatic feed from the wearable devices, and building that feed is the main engineering step that remains.

On intellectual property, I wrote the PULSE research pipeline and application. The combined CareSense product contains the work of four members, so its ownership would have to follow the policy of SLIIT and an agreement among the group. The main libraries used, including LightGBM, SHAP, FastAPI and React, are released under permissive open-source licenses, but the terms should be checked again before any release. The TILES-2018 data were given to this project under a signed data use agreement for research, shown in Appendix D. A commercial version should be

retrained on data collected from the care homes themselves, or the data provider should be asked for permission.

On compliance, Sri Lanka's Personal Data Protection Act would apply to any deployment that handles caregivers' personal data [30]. Staff would have to be told what is monitored and would have to agree, a lawful basis would be needed, and an ethics review would come first, as set out in Section 3.6.

8.5 Contribution to the Sustainable Development Goals

This research aligns with Sustainable Development Goal (SDG) 3, Good Health and Well-being [31], by addressing caregiver workforce sustainability, which is essential for delivering safe, high-quality healthcare services. A stable and well-supported caregiver workforce directly contributes to improved patient outcomes, reduced errors, and overall system resilience [6].

At the same time, the research aligns with the Centre of Excellence for Artificial Intelligence (CEAI) by applying advanced AI and data science techniques to solve a critical real-world healthcare problem. The proposed system integrates machine learning, temporal modeling, graph-based relational analysis, and explainable AI within a practical web-based solution, demonstrating how intelligent technologies can be used to enhance workforce management, support decision-making, and improve healthcare delivery in a scalable and impactful manner.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research set out to design, build and evaluate PULSE, the stress estimation and workload redistribution module of CareSense, using wearable and survey records from hospital workers. The research pipeline separates participants before any feature is built, learns the heart rate expected for each caregiver in a first stage, and estimates the probability of elevated self-reported stress in a second. When it was tested on 27 caregivers who took no part in training, the pipeline reached a ROC AUC of 0.7211, with a 95% interval of 0.6644 to 0.7699. A repeated, participant-grouped validation gave a mean of 0.6991, which supports the held-out result.

The experiments also show where the predictive signal comes from, and this defines the claim the evidence supports. The previous stress response carries most of the signal. The full model ranked assessments better than that response alone by 0.0247, although the interval for this difference includes zero, and the wearable features performed close to chance when survey history was removed. Proximity centrality made no measurable difference in three model families. PULSE is therefore best described as a transparent prototype that ranks elevated reported stress at a moderate level.

On the product side, PULSE operates as a complete working system. A supervisor can view tiered estimates, factor explanations, histories of sustained risk and a same-unit suggestion that stays open to human review, all behind role-based access. The module is integrated with the other three CareSense components through shared authentication, navigation, audit and model-registry services.

9.2 Achievement of Objectives

Table 9.1 returns to the six sub-objectives set out in Section 1.5 and summarizes how each of them was achieved.

Table 9.1: Status of the original objectives

Objective	Status	Evidence and caveat
1. Build workload and behavioral features	Achieved	Features from heart rate, steps, sleep and survey history, built at each survey cut-off (Section 3.3.1)
2. Create personal baselines and deviation scores	Achieved	Historical deviations and a learned expected heart rate (Section 3.3.2), which raised the test ROC AUC from 0.7086 to 0.7211
3. Add temporal detection over successive assessments	Achieved	Previous-response feature in the model and a sustained-risk rule in the application (Section 4.4.4); evaluating the rule against observed deterioration is a task for later work
4. Examine relational structure from proximity data	Achieved through investigation	Matched comparison in three model families, which found no measurable gain from proximity centrality (Section 6.1.6)
5. Explain individual risk assessments	Achieved	SHAP factor lists in the application (Sections 6.1.7 and 6.2), shown to reviewers in the walkthrough (Section 6.3)
6. Design and demonstrate rule-based, human-reviewed redistribution	Achieved through demonstration	Rule and interface implemented (Section 4.4.4) and demonstrated on the historical readings of the demonstration group

The research questions in Section 1.6 were answered as Table 6.7 summarizes. RQ1 and RQ4 received clear answers within the cohort. RQ2 and RQ3 produced negative or inconclusive findings, which are reported as results in their own right. RQ5 was answered through the working application and the reviewer walkthrough.

9.3 Individual Research and Product Contributions

Within the CareSense project, my component is the one that turns a prediction into a staffing suggestion. The research contribution is a documented pipeline that was evaluated conservatively and that reports its negative findings alongside the positive ones. The product contribution is a full-stack module, with its own database rules, service, interface and redistribution logic, integrated into the shared platform. Table 1.3 lists these areas in detail. The group thesis will place these contributions beside those of the other three components.

9.4 Recommendations and Future Directions

The recommendations below follow from the limitations discussed in Chapter 7. I have listed them in the order in which I would carry them out.

1. Extend the usability evaluation beyond the initial walkthrough, using permitted reviewers and authorized examples, with the tasks and measures set out in Section 6.3.
2. Replace the single-seed allocation with several seeds, and set aside a test group that is not inspected during development.
3. Test the model on data from care-home staff, with ethics approval and data-protection safeguards, before making any claim about that population.
4. Improve the redistribution rule with skill mix, minimum staffing and capacity limits, and ask a domain expert to judge the suggestions.
5. Revisit proximity with better network coverage, and test whether a different construction of the network changes the result.

If these steps support the approach, a prospective pilot with consent and clear governance would allow the effect of the module on supervisors' decisions to be measured.

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GLOSSARY

Table 9.2: Terms used in this thesis

Term	Meaning
Elevated stress	In this research, a self-reported stress rating of 3 or above on a 1 to 5 scale. It is a research label and not a diagnosis.
Ecological momentary assessment	A short self-report collected close to the moment of experience in everyday settings.
Expected heart rate	The heart rate that the first-stage model predicts for a person from their recent history and the day of the week.
Heart-rate deviation	The difference between observed and expected heart rate, used as an input to the classifier.
Held-out group	People set aside before features were built, used only to test the final model.
Grouped validation	Validation in which all rows from one person fall in the same fold.
Lag feature	The previous usable stress response of the same person.
Proximity centrality	A measure of how connected a person is in a network built from Bluetooth contacts.
SHAP value	A per-feature contribution to a single prediction, based on Shapley values.
Sustained risk	In the application, at least three of the last five scored assessments predicted as elevated.
Unscored day	A day in the replay for which no score is shown because too little history exists.

APPENDICES

Appendix A Participant allocation and usable data

Table A.1 gives the participant allocation and the number of usable rows at each stage. The allocation used a fixed random seed of 42 and was made before any feature was built.

Table A.1: Participant allocation and usable survey rows

Group	Allocated people	Usable people	Usable rows	Rows after Stage 1 filter
Training	140	132	5,989	5,977
Validation	30	30	1,296	1,292
Test	27	27	1,407	1,406
Demonstration	15	not used in modelling	not applicable	not applicable
Total	212	189 (modelling)	8,692	8,675

The 15 demonstration people were kept apart from all modelling and supply the historical readings used in the application’s replay.

Appendix B Disclosure of AI assistance

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for
3. my own understanding and work.
4. I agree to use AI tools responsibly, ethically, and transparently.
5. I will clearly disclose any use of AI tools in my submissions.
6. I understand that I bear full responsibility and accountability for any content, code,
7. model, or output generated using AI tools.
8. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
9. I understand that failure to comply with this policy may result in academic penalties.

In line with this declaration, I used AI tools as a support tool during the preparation of this thesis. It helped me to organize the structure against the faculty template, to draft and edit text from my own project notes, to format the document and to check the details of the cited references. The research design, the experiments, the interpretation of the results and the conclusions are my own. I reviewed the text before submission, and I take full responsibility for its content, its claims and its citations. I am able to explain and defend each of them.

Appendix C Log Book Records for Visitations

Figure C.1 to Figure C.3 reproduce the records of the consultations that shaped the requirements of the module.

07.04.2026	Conducted a group interview with Mr. Vidyantha Vasantha Abeyrathne, the founder of Remark Elder Care (a subsidiary of Feathered Health Care), and visited the care home to gather real-world caregiver requirements. Based on the insights and meeting notes, I designed a workflow to predict caregiver stress using periodic voice logs and proposed a solution for workload redistribution through substitute caregivers.

Figure C.1: Consultation record 1 – screenshot from log book

21.04.2026	Visited CCHS (Colombo Caregiver & Health Science School) to gain a practical understanding of real-world caregiving workflows and documentation practices. The purpose of the visit was to observe how caregivers are trained, how they manage daily responsibilities, and how they record patient-related information. Special attention was given to understanding the logging process maintained by students, particularly how they document the condition, behavior, and daily care activities of the individuals they are assigned to.

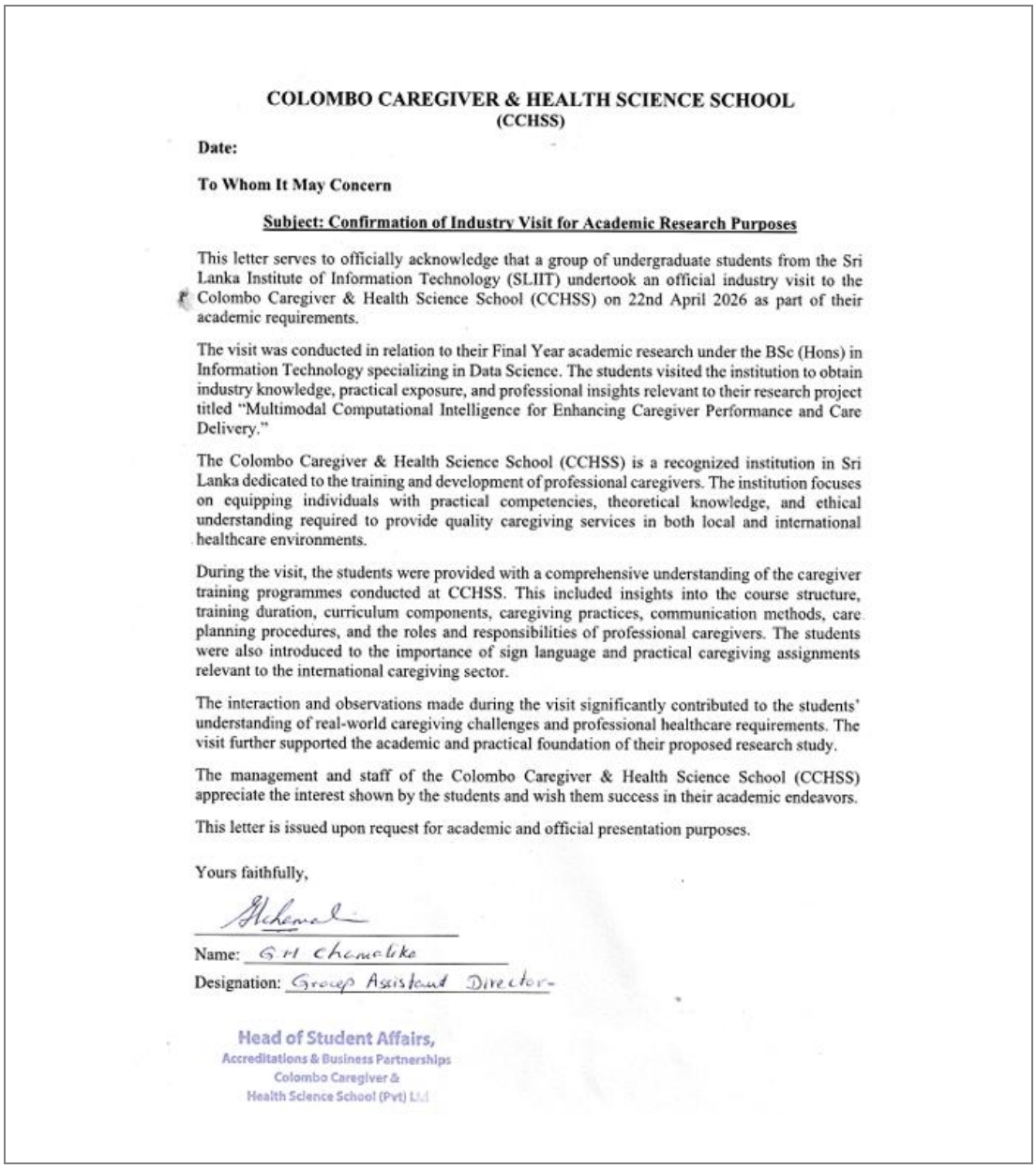
Figure C.2: Consultation record 2 – screenshot from log book

22.04.2026	Visited a local elder care home *"Love and Care Home"* on two occasions to closely observe real world caregiving practices and documentation methods. The visits focused on understanding how caregivers manage different types of patients and how daily logs are maintained in practical settings. Interactions were held with the director and caregiving staff to gain deeper insights into operational workflows and patient-specific care routines. It was observed that log records vary significantly depending on patient conditions; for example, kidney patients require detailed monitoring of fluid intake, output levels, and dietary restrictions, while other patients may require logs related to mobility, medication schedules, or behavioral changes.

Figure C.3: Consultation record 3 – screenshot from log book

Appendix D Signed documents and approvals

This appendix holds the signed documents that support the work reported in this thesis. They cover the approval of the thesis for submission, the visits that informed the requirements of the module, the stakeholder walkthrough described in Section 6.3, and the agreement under which the TILES-2018 data were released to the project.



COLOMBO CAREGIVER & HEALTH SCIENCE SCHOOL
(CCHSS)

Date:

To Whom It May Concern

Subject: Confirmation of Industry Visit for Academic Research Purposes

This letter serves to officially acknowledge that a group of undergraduate students from the Sri Lanka Institute of Information Technology (SLIIT) undertook an official industry visit to the Colombo Caregiver & Health Science School (CCHSS) on 22nd April 2026 as part of their academic requirements.

The visit was conducted in relation to their Final Year academic research under the BSc (Hons) in Information Technology specializing in Data Science. The students visited the institution to obtain industry knowledge, practical exposure, and professional insights relevant to their research project titled "Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery."

The Colombo Caregiver & Health Science School (CCHSS) is a recognized institution in Sri Lanka dedicated to the training and development of professional caregivers. The institution focuses on equipping individuals with practical competencies, theoretical knowledge, and ethical understanding required to provide quality caregiving services in both local and international healthcare environments.

During the visit, the students were provided with a comprehensive understanding of the caregiver training programmes conducted at CCHSS. This included insights into the course structure, training duration, curriculum components, caregiving practices, communication methods, care planning procedures, and the roles and responsibilities of professional caregivers. The students were also introduced to the importance of sign language and practical caregiving assignments relevant to the international caregiving sector.

The interaction and observations made during the visit significantly contributed to the students' understanding of real-world caregiving challenges and professional healthcare requirements. The visit further supported the academic and practical foundation of their proposed research study.

The management and staff of the Colombo Caregiver & Health Science School (CCHSS) appreciate the interest shown by the students and wish them success in their academic endeavors.

This letter is issued upon request for academic and official presentation purposes.

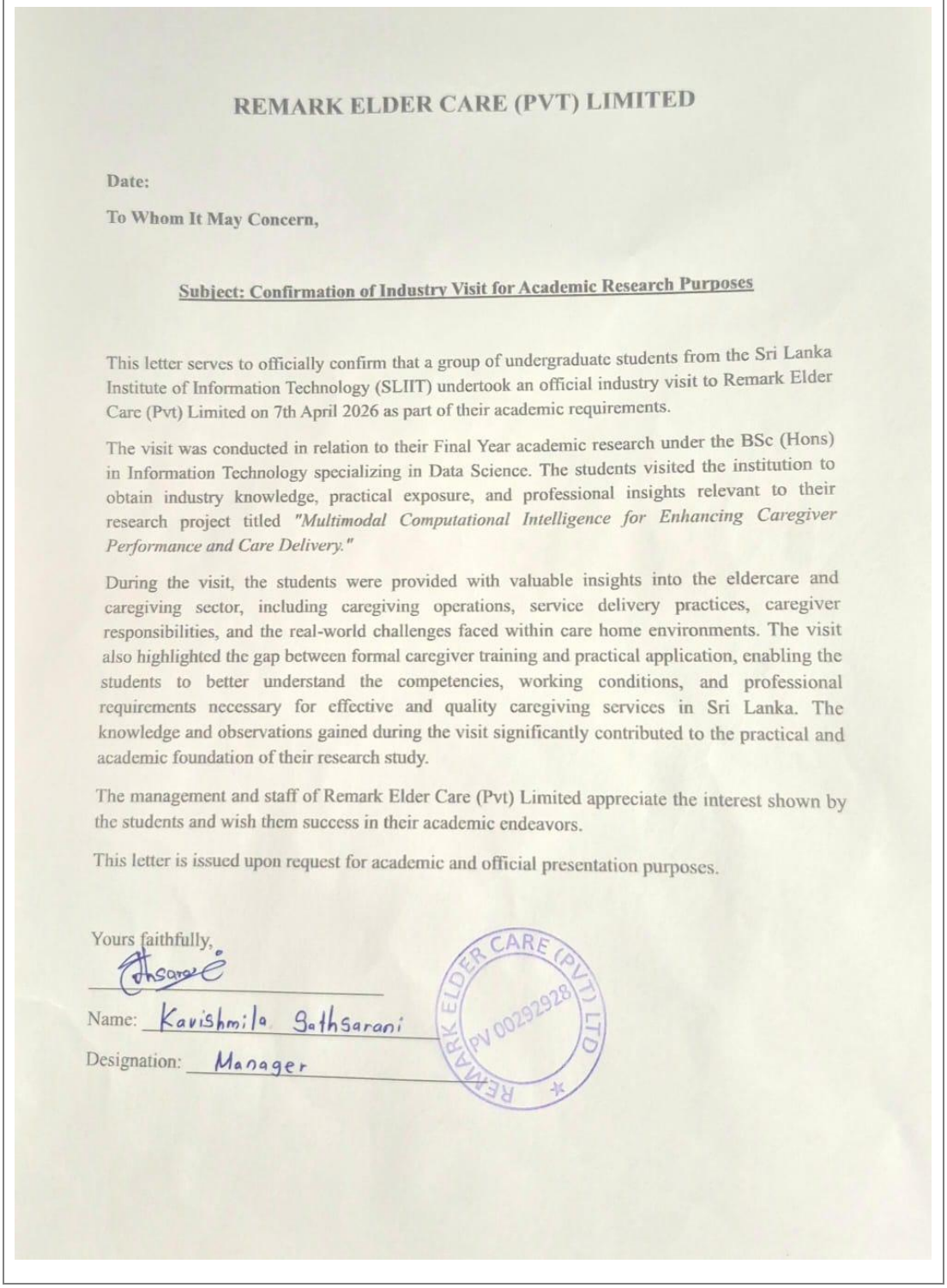
Yours faithfully,

Name: G H Chemeleke

Designation: Group Assistant Director-

Head of Student Affairs,
Accreditations & Business Partnerships
Colombo Caregiver &
Health Science School (Pvt) Ltd

Figure D.1: Signed confirmation of the group visit to the Colombo Caregiver & Health Science School



REMARK ELDER CARE (PVT) LIMITED

Date:

To Whom It May Concern,

Subject: Confirmation of Industry Visit for Academic Research Purposes

This letter serves to officially confirm that a group of undergraduate students from the Sri Lanka Institute of Information Technology (SLIIT) undertook an official industry visit to Remark Elder Care (Pvt) Limited on 7th April 2026 as part of their academic requirements.

The visit was conducted in relation to their Final Year academic research under the BSc (Hons) in Information Technology specializing in Data Science. The students visited the institution to obtain industry knowledge, practical exposure, and professional insights relevant to their research project titled *"Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery."*

During the visit, the students were provided with valuable insights into the eldercare and caregiving sector, including caregiving operations, service delivery practices, caregiver responsibilities, and the real-world challenges faced within care home environments. The visit also highlighted the gap between formal caregiver training and practical application, enabling the students to better understand the competencies, working conditions, and professional requirements necessary for effective and quality caregiving services in Sri Lanka. The knowledge and observations gained during the visit significantly contributed to the practical and academic foundation of their research study.

The management and staff of Remark Elder Care (Pvt) Limited appreciate the interest shown by the students and wish them success in their academic endeavors.

This letter is issued upon request for academic and official presentation purposes.

Yours faithfully,

Name: Kavishmila Sathsarani

Designation: Manager

REMARK ELDER CARE (PVT) LTD PV 00292928

Figure D.2: Signed record of the stakeholder walkthrough with the representative of Remark Elder Care (Section 6.3)

Recipient INFORMATION and AUTHORIZED SIGNATURE

Recipient Scientist: Prof. Samantha Thelijjagoda <Please complete>

Recipient Organization: Sri Lanka Institue of Information Technology <Please complete>

Digitally signed by Samantha Thelijjagoda
Date: 2026.07.16 09:13:57 +09'00'

Signature of Authorized Official Date

Title:

Address all correspondence related to this agreement to:

Prof. Samantha Thelijjagoda, Senior Professor [insert address for notices]
Sri Lanka Institute of Information Technology (SLIIT), New Kandy Road, Malabe, 10115, Sri Lanka.
Researcher: Dayana Priyadharshani Kumar - it22178640@my.sliit.lk

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USC INFORMATION and AUTHORIZED SIGNATURE

USC Principal Investigator: Dr. Shrikanth Narayanan

07/16/2026
Date

Judy Genovese
Sr. Transactions Officer
USC Stevens Center for Innovation

Address all correspondence related to this agreement to:

USC Stevens Center for Innovation
1150 South Olive Street, Suite 2300
Los Angeles, CA 90015

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Figure D.3: Signature page of the TILES-2018 data use agreement

Source: Project records.

Appendix E Notebook screenshots

The screenshots in this appendix reproduce the outputs of the final research notebook on which the results in Chapter 6 are based. Each caption names the notebook section from which the output was taken and the section of the thesis that it supports. The notebook is organized in two parts, Part 1 for data preparation and Part 2 for modelling and analysis, and its sections are mapped to the thesis in Appendix F.

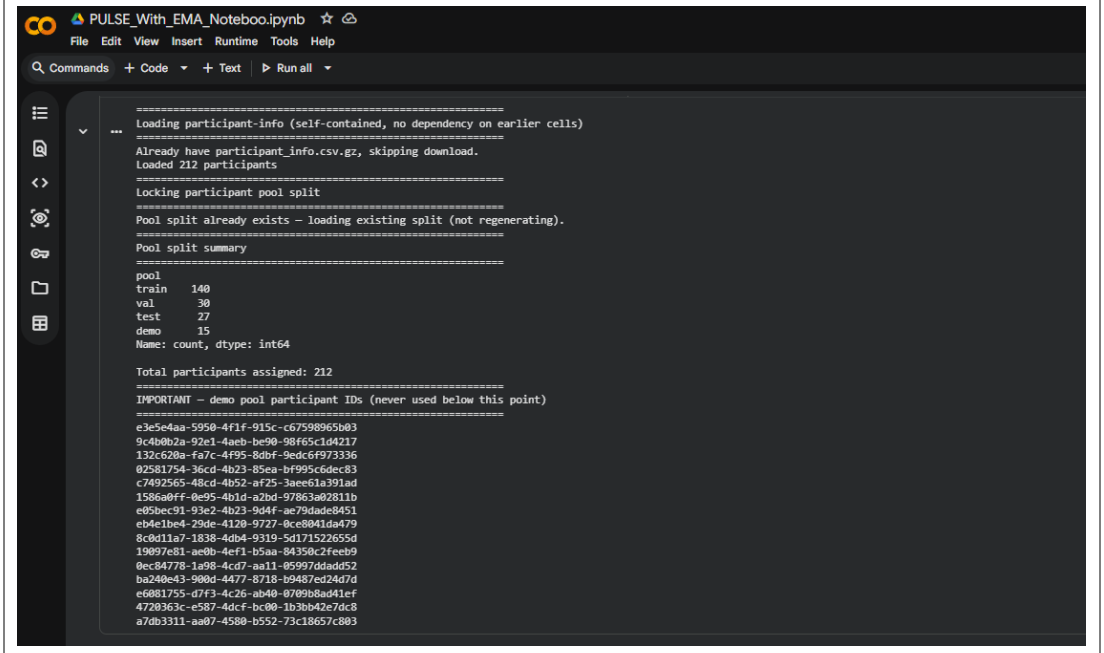


Figure E.1: Participant allocation into training, validation, test and demonstration groups (notebook section 1.2; Section 3.2.3 and Appendix A)

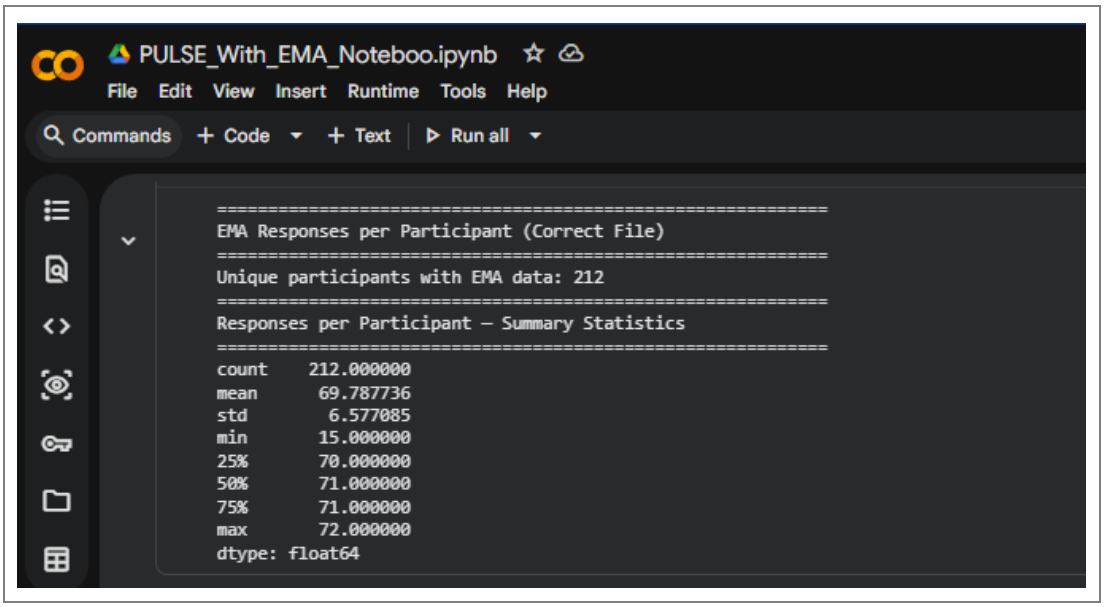


Figure E.2: Completeness and distribution of the stress ratings in the EMA file (notebook section 1.1; Section 3.2.2)

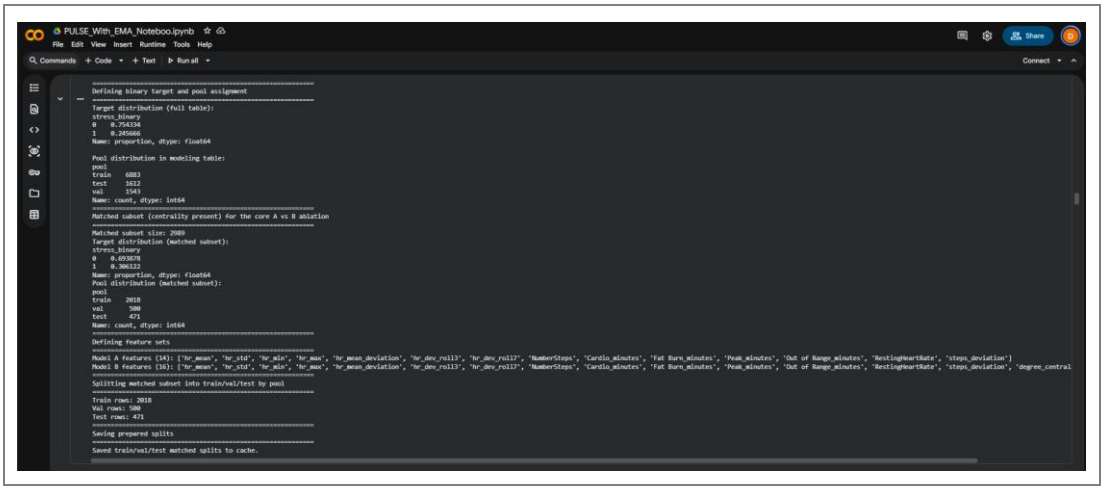


Figure E.3: Target distribution and group sizes of the modelling table (notebook section 1.7; Sections 3.2.2 and 3.3.4)

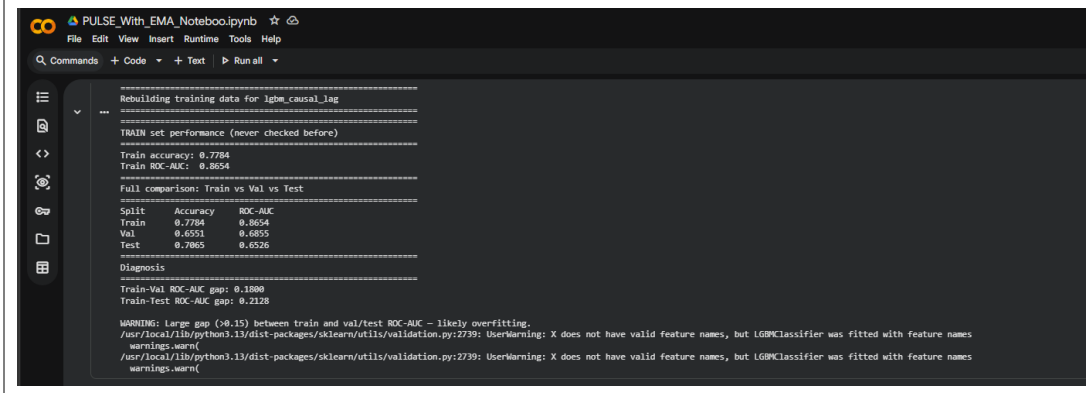


Figure E.4: Overfitting check comparing training, validation and test performance (notebook section D.4; Section 6.1.1)

Source: Author's notebook outputs.

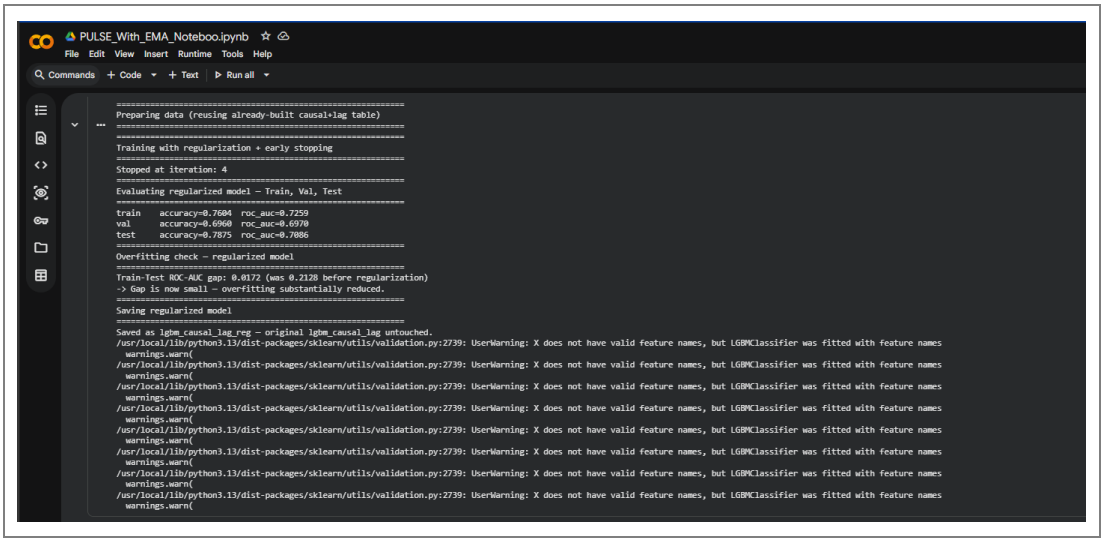


Figure E.5: Regularised model with early stopping (notebook section D.5; Section 6.1.1)

Source: Author's notebook outputs.

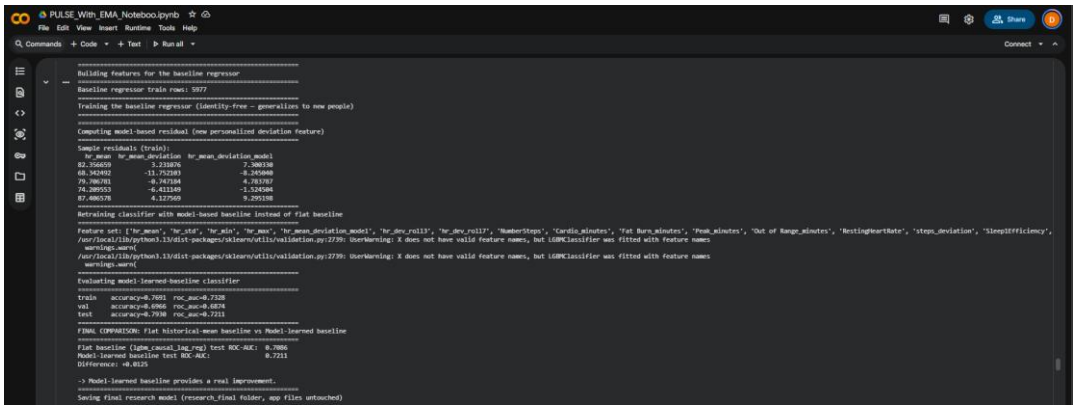


Figure E.6: Feature set of the final model and the learned heart-rate baseline (notebook section D.6; Sections 3.3.2 and 3.3.3)

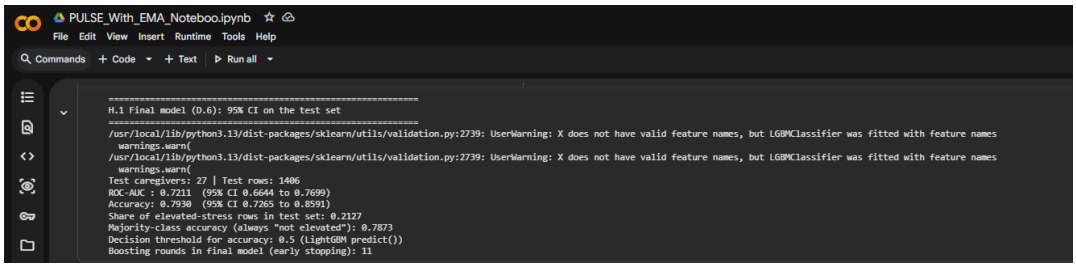


Figure E.7: Confidence intervals for the final model on the test group (notebook section H.1; Sections 6.1.2 and 6.1.3)

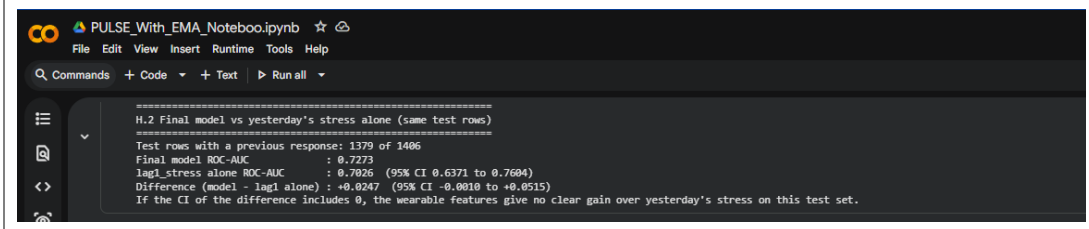


Figure E.8: Final model compared with the previous stress response alone (notebook section H.2; Section 6.1.4)

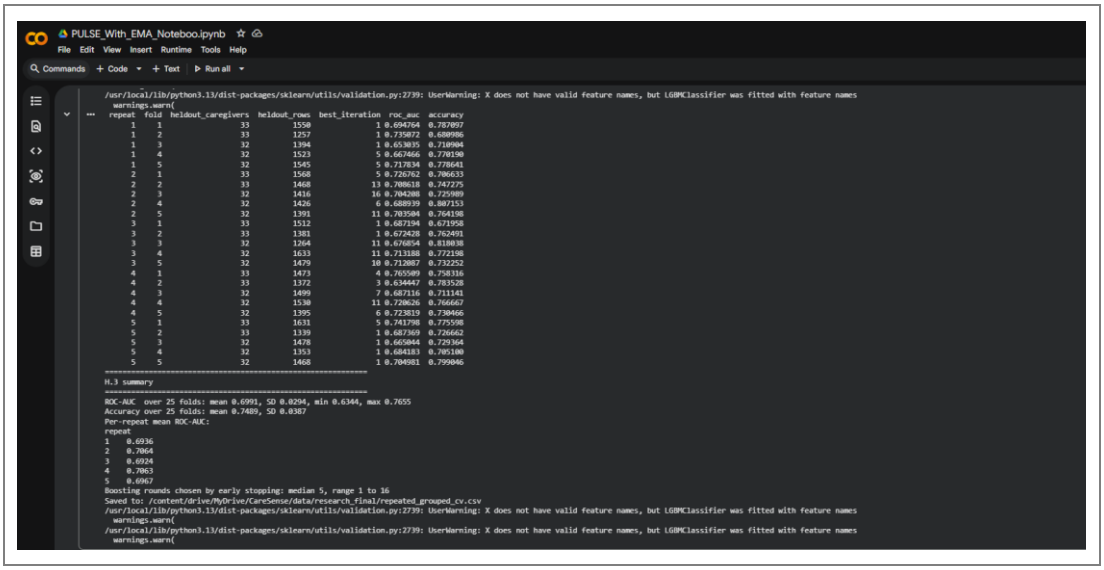


Figure E.9: Summary of the repeated participant-grouped cross-validation (notebook section H.3; Section 6.1.3)

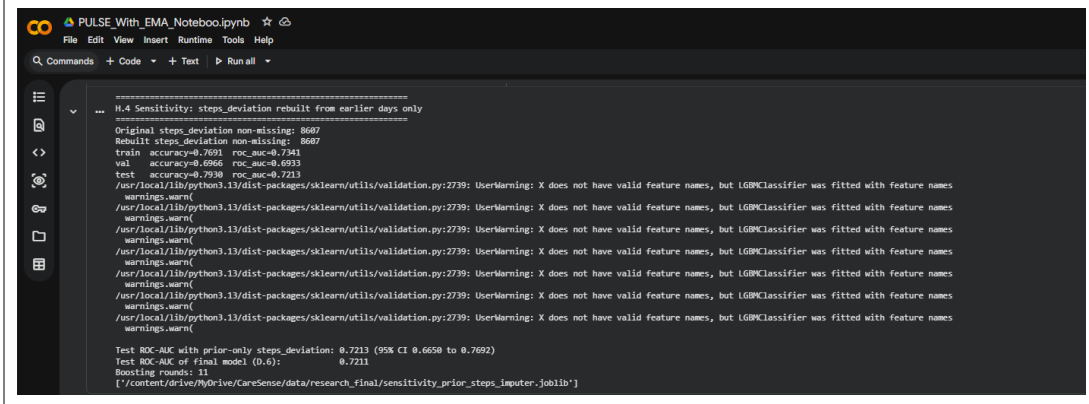


Figure E.10: Sensitivity check with the steps deviation rebuilt from earlier days only (notebook section H.4; Section 6.1.5)

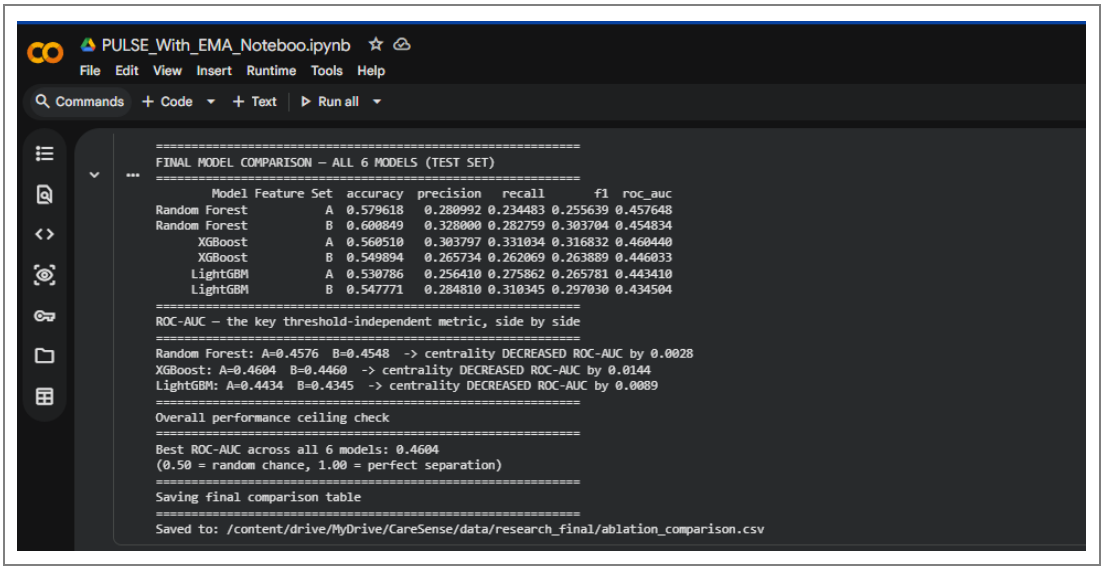


Figure E.11: Centrality ablation across the six models of the same-day experiment (notebook section B.4; Section 6.1.6)

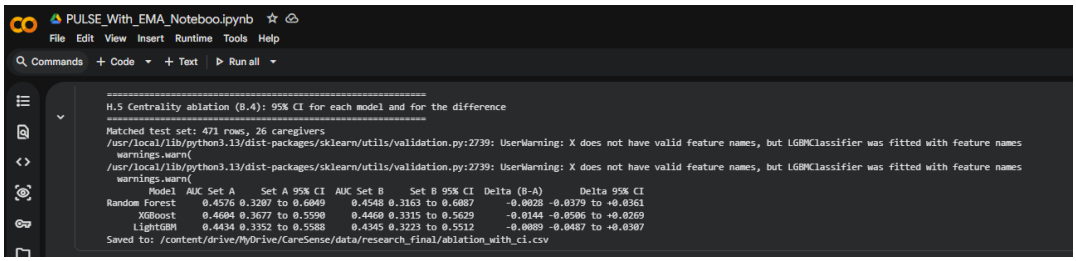


Figure E.12: Confidence intervals for the centrality ablation (notebook section H.5; Section 6.1.6)



Figure E.13: Mean absolute SHAP values of the final model (notebook section D.7; Section 6.1.7)

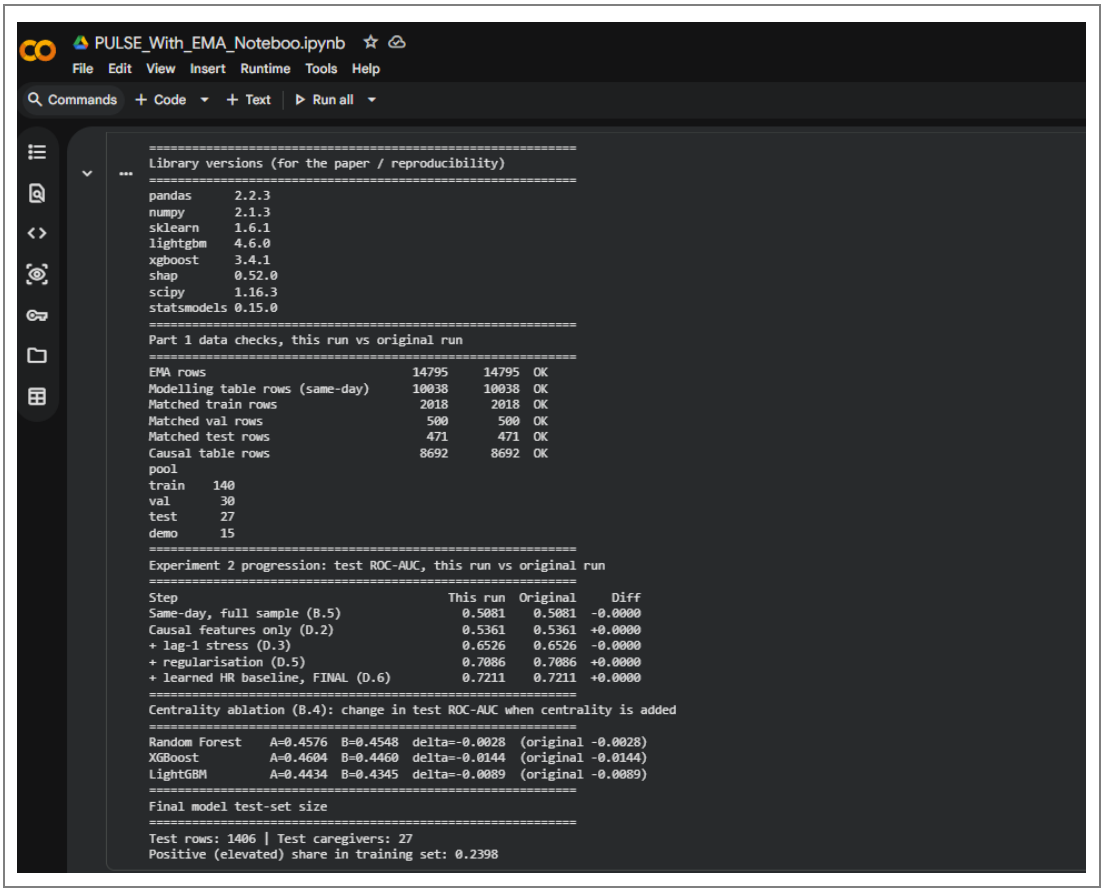


Figure E.14: Library versions and comparison of this run with the original run (notebook section E; Section 3.7)

Appendix F Notebook sections and their use in the thesis

Table F.1 lists the sections of the final research notebook and the sections of the thesis in which their results are used. It allows each reported value to be traced to the output that produced it.

Table F.1: Notebook sections and the thesis sections that use them

Notebook section	Content	Thesis section
1.1	EMA responses: shape, completeness and date range	3.2.2
1.2	Locked participant allocation with seed 42	3.2.3, Appendix A
1.3	Same-day daily feature table	3.3.1
1.4 to 1.6	Badge files, co-location network and merged centrality features	3.3.4
1.7	Target, matched sample and splits	3.2.2, 3.3.4
B	Same-day features and the centrality ablation in three model families	6.1.6
C	Statistical analyses on the same-day table	6.1.8
D.1 to D.5	Causal features, previous stress response, overfitting check and regularisation	3.3.1, 6.1.1
D.6	Final model with the learned heart-rate baseline	3.3.2, 3.3.3, 6.1.2
D.7	SHAP explanation of the final model	6.1.7
E	Reproducibility check against the original run	3.7
H.1 to H.5	Confidence intervals, comparison with the previous response alone, repeated grouped validation, prior-only steps baseline and centrality intervals	6.1.2 to 6.1.6